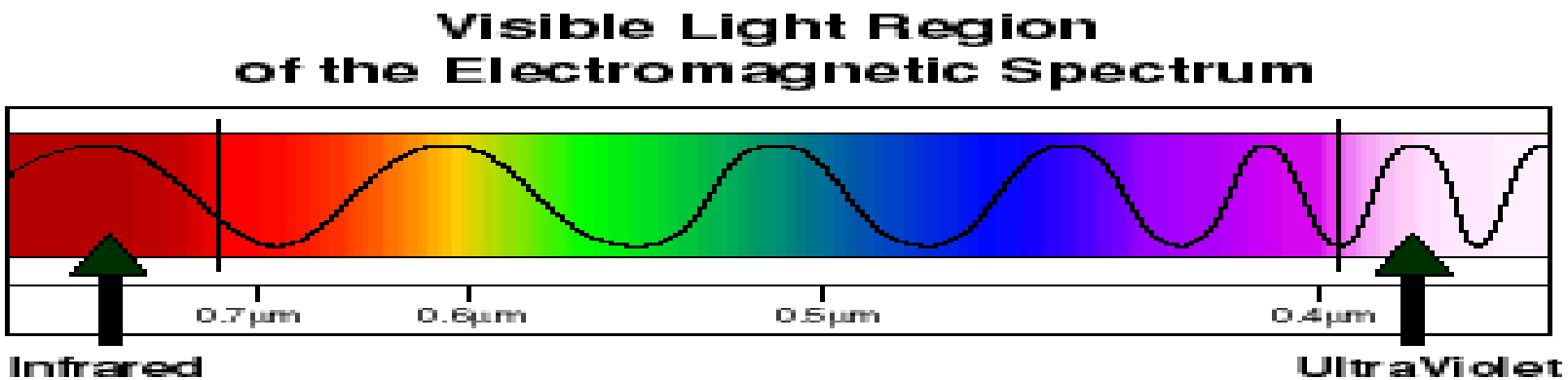
OPTICAL NETWORKING

WDM and DWDM

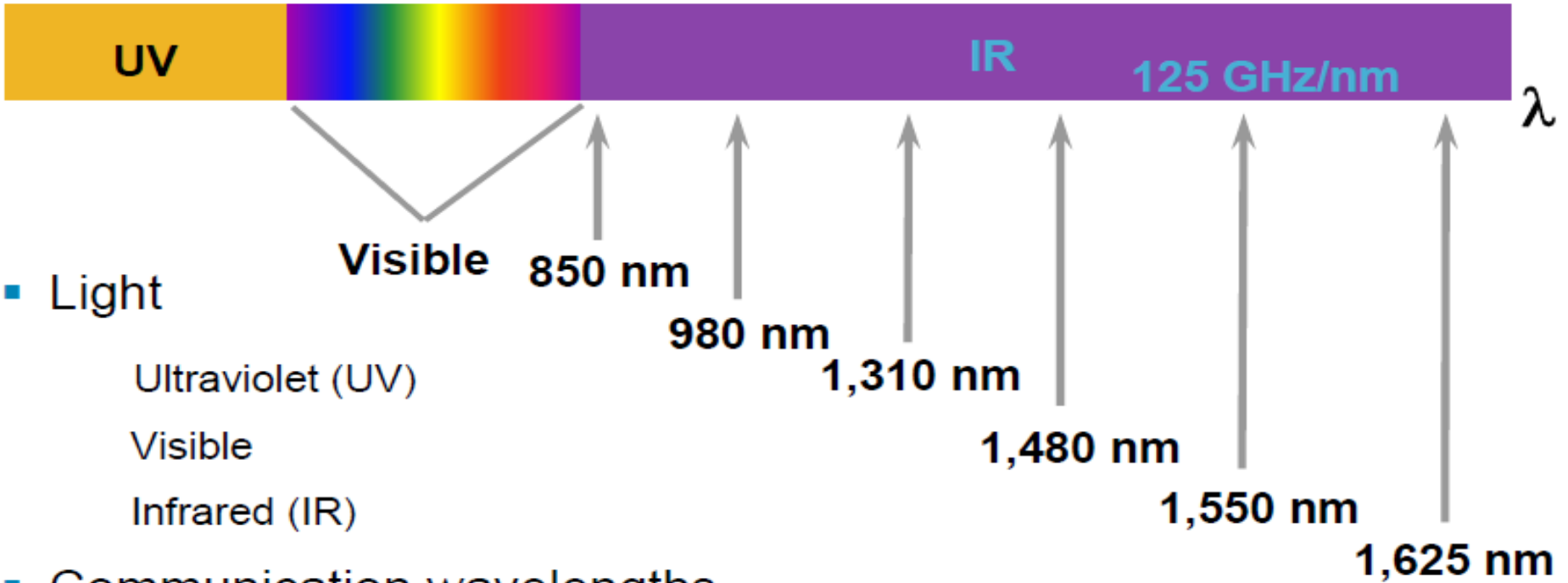
รศ.ดร.ชวลิต เบญจางคประเสริฐ

Optical Fundamentals

- Wavelength (λ): length of a wave and is measured in nanometers, 10^{-9}m (nm)
  - **400 nm (violet) to 700 nm (red) is visible light**
  - **Fiber optics primarily use 850, 1310, & 1550 nm**
- Frequency (*f*): measured in TeraHertz, 10^{12} (THz)
- Speed of light = 3×10^{8} m/sec



Optical Spectrum

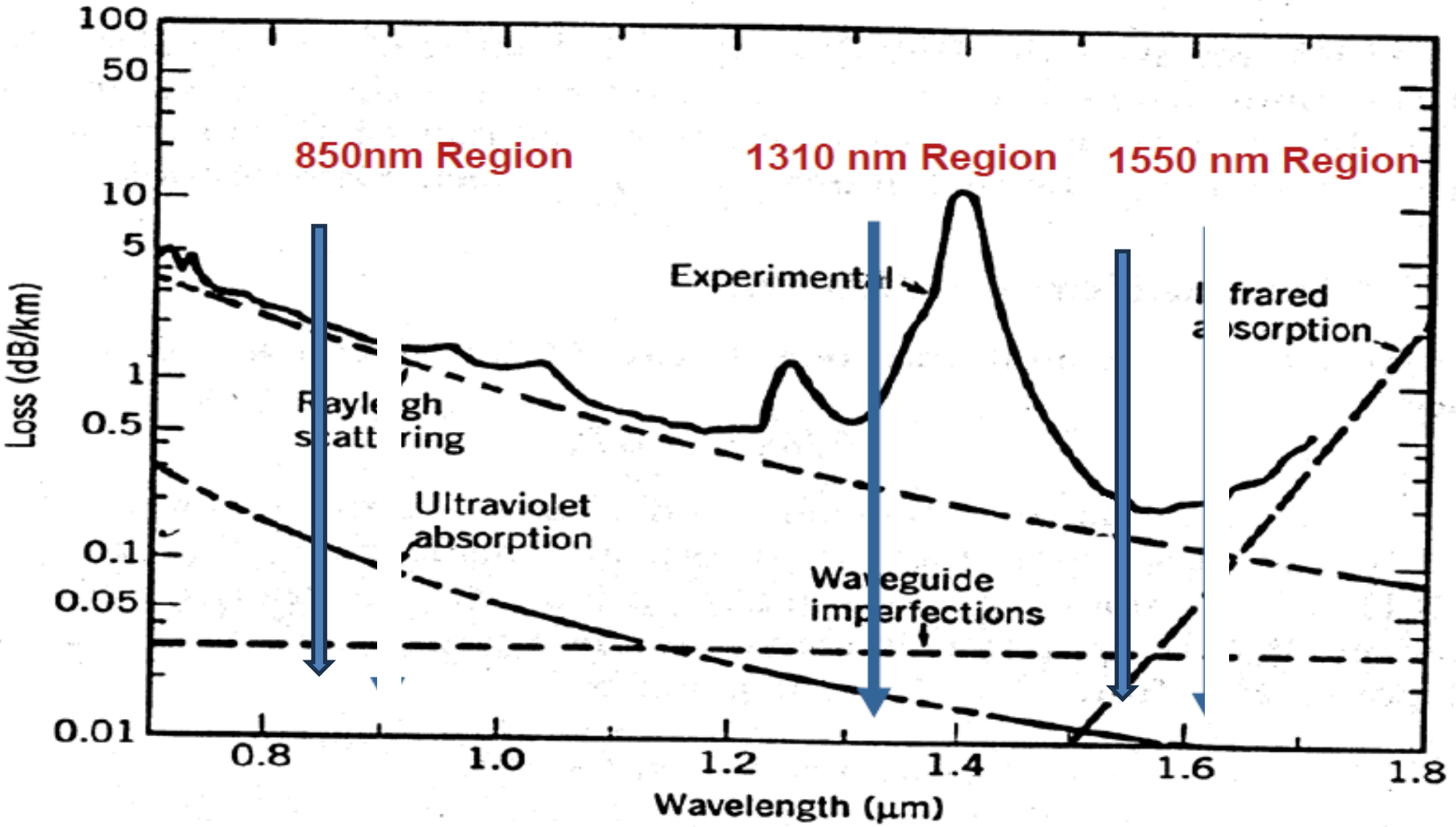


- Light
  - Ultraviolet (UV)
  - Visible
  - Infrared (IR)
- Communication wavelengths
  - 850, 1310, 1550 nm
  - Low-loss wavelengths
- Specialty wavelengths
  - 980, 1480, 1625 nm

1550nm	193,548.4GHz
1551nm	193,424.6GHz
1nm	125 GHz

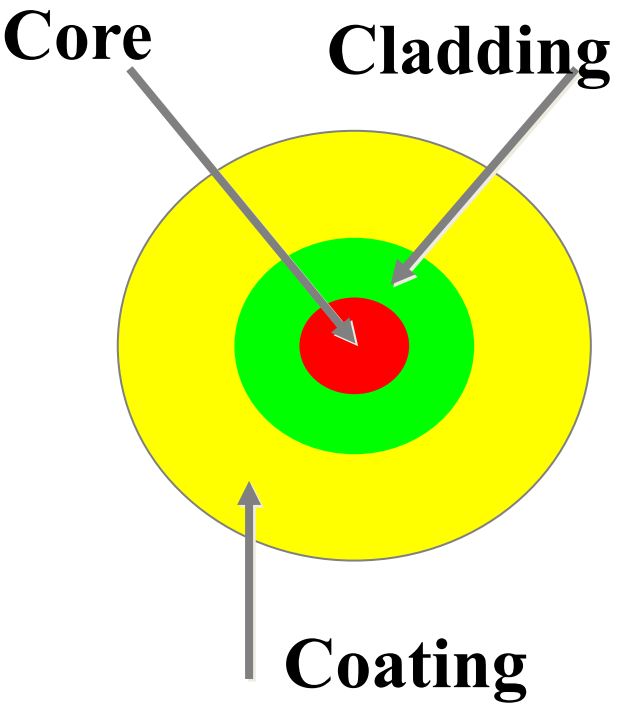
Bandwidth

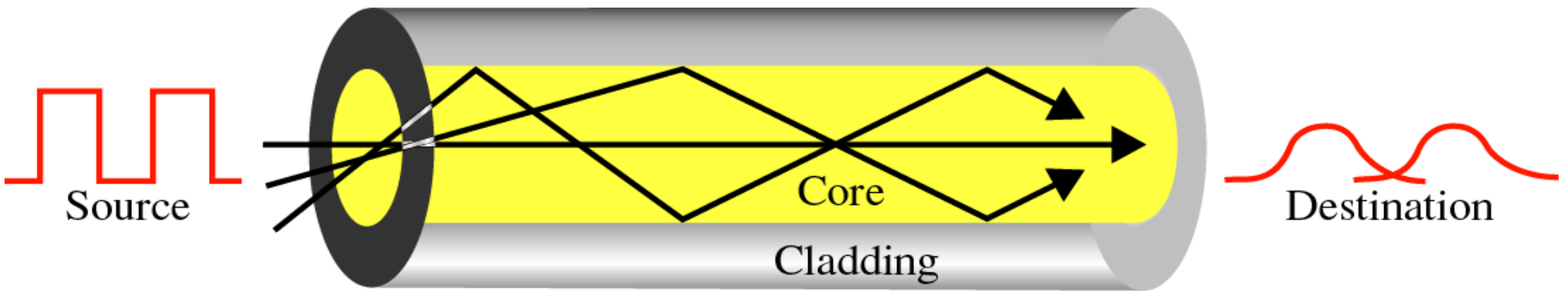
Attenuation Response at Different Wavelengths



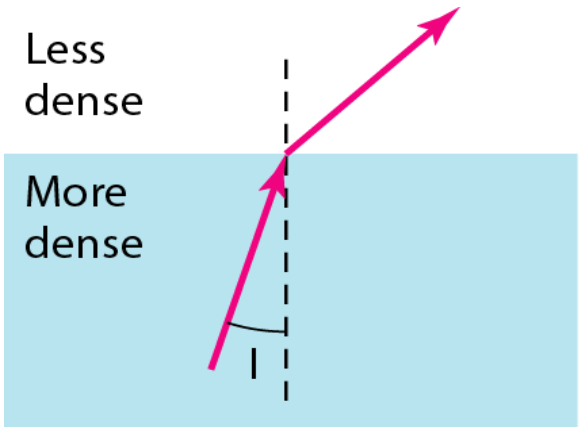
Optical Fiber

- An optical fiber is made of three sections:
  - The core carries the light signals
  - The cladding keeps the light in the core
  - The coating protects the glass

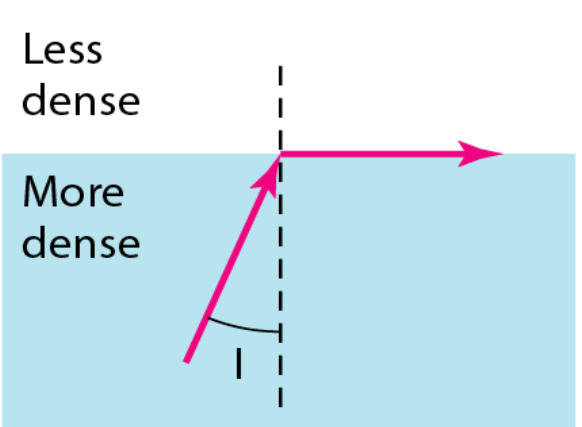




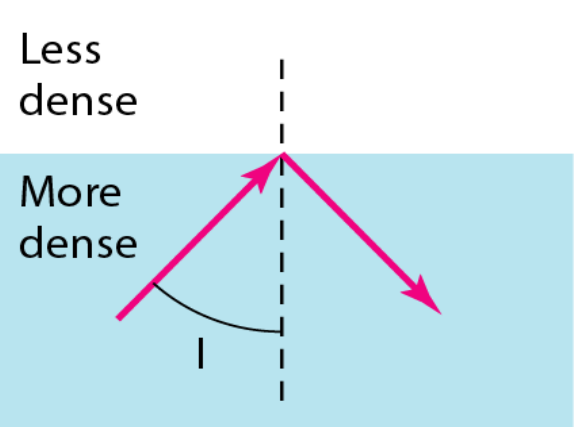
Bending of light ray



I < critical angle,
refraction

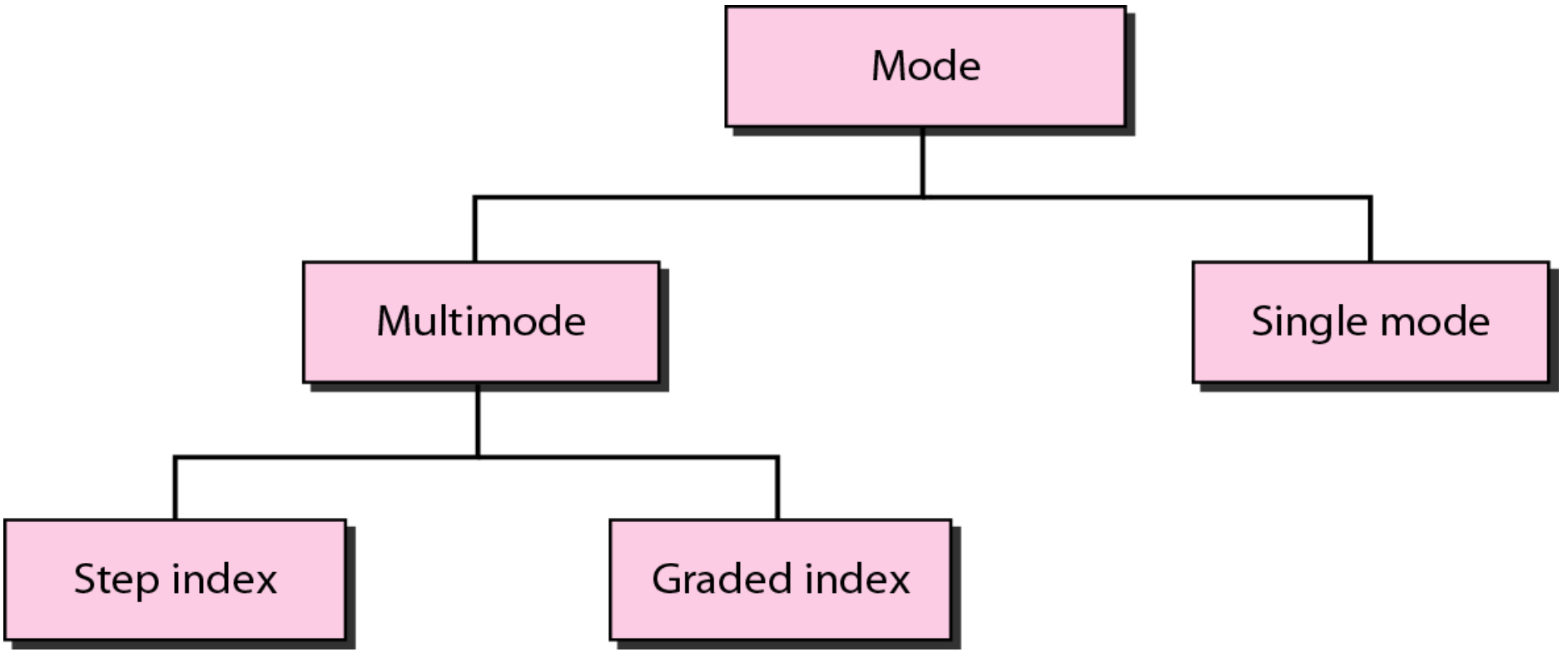


I = critical angle,
refraction

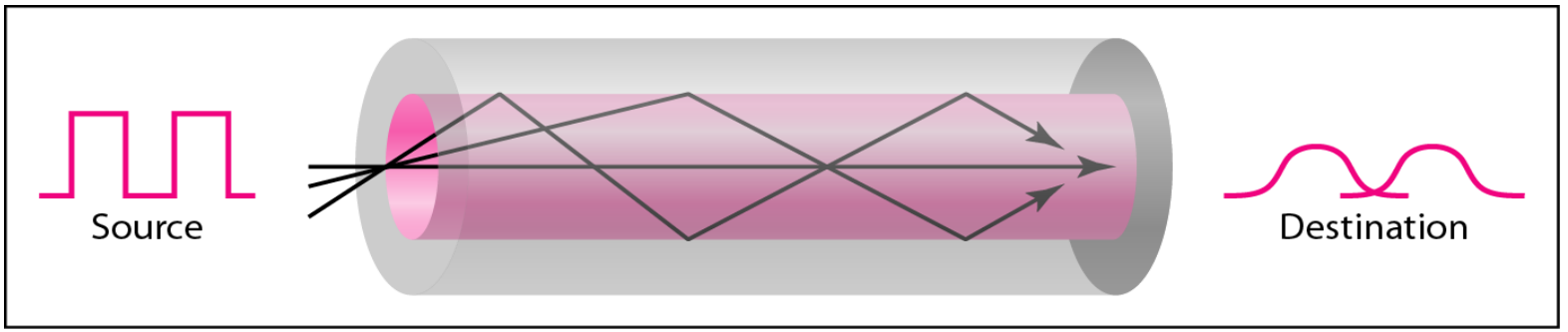


I > critical angle,
reflection

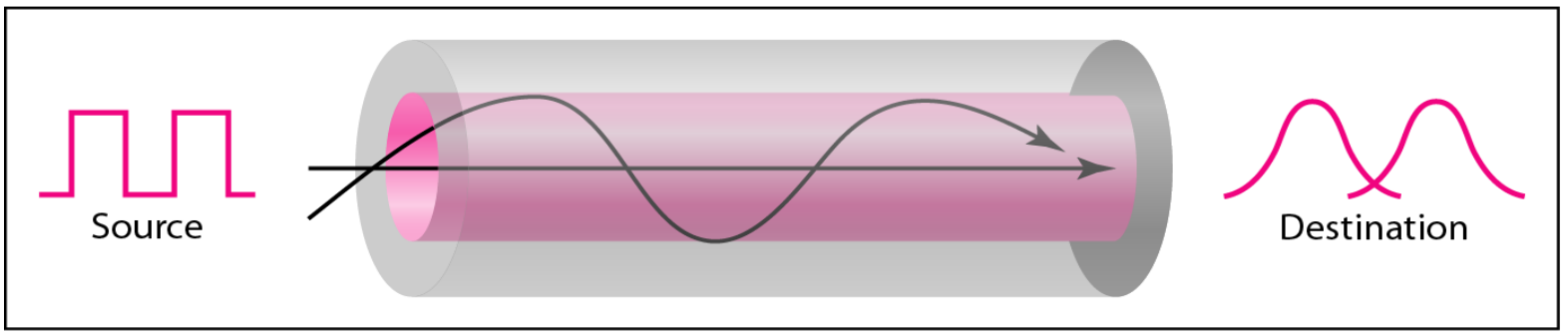
Propagation modes



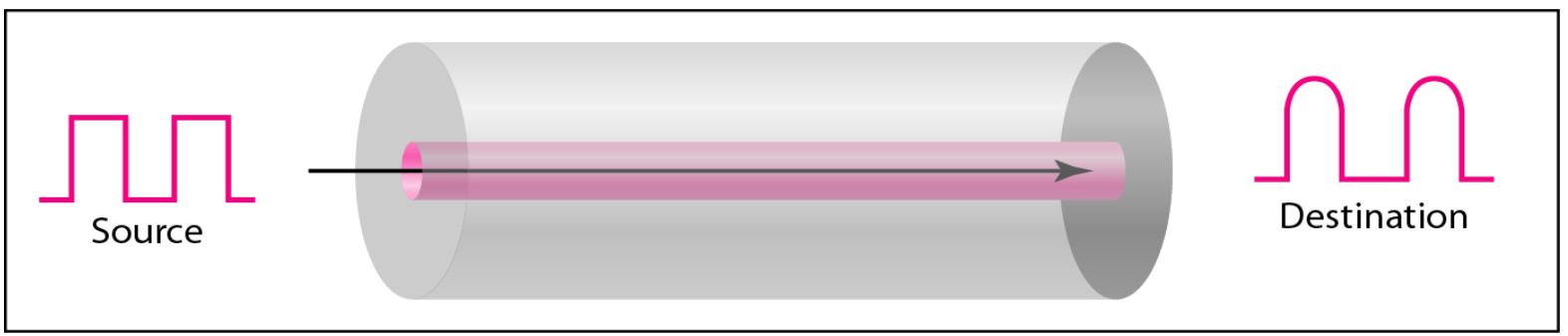
Modes



a. Multimode, step index



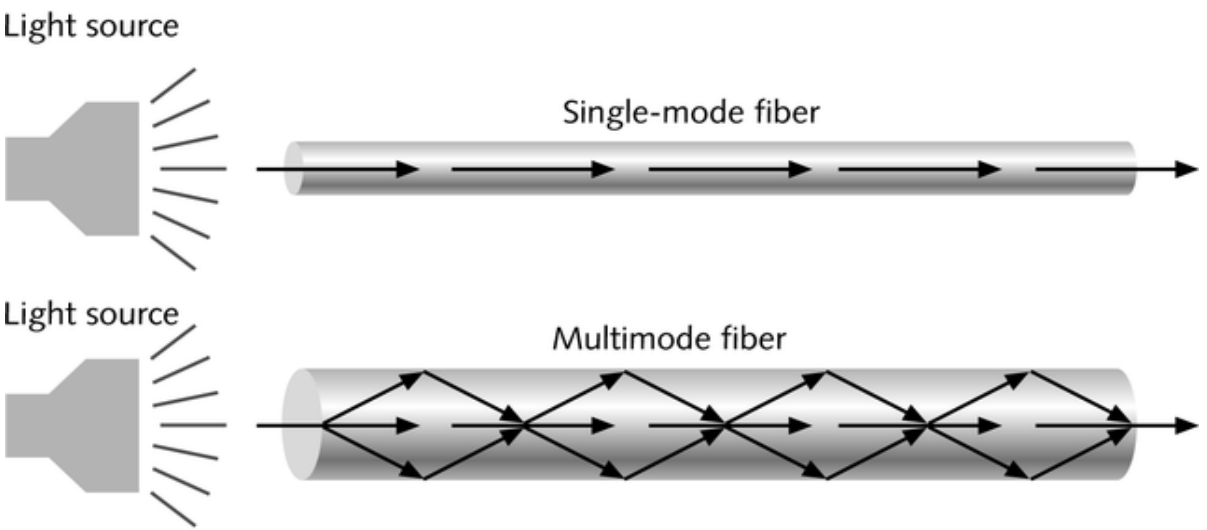
b. Multimode, graded index



c. Single mode

Optical Fiber (cont.)

- Single-mode fiber
  - Carries light pulses by laser along single path.
- Multimode fiber
  - Many pulses of light generated by LED travel at different angles.



SM: core=8-10 μm cladding=125 μm

MM: core=50 or 62.5 μm cladding=125 μm

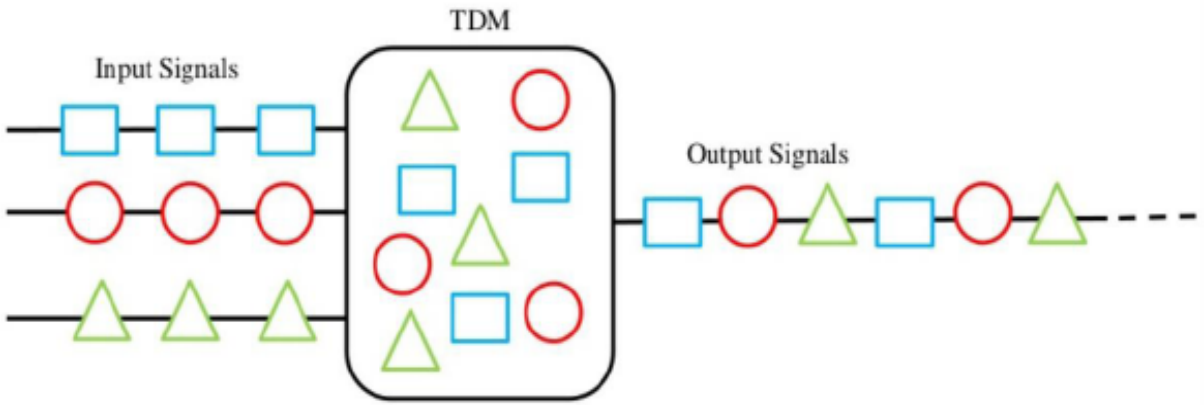
Multiplexing

- Multiplexing
  - a process where multiple analog message signals or digital data streams are combined into one signal over a shared medium.
- Types
  - Time division multiplexing (TDM).
  - Frequency division multiplexing (FDM).
- Optically
  - Time division multiplexing (TDM).
  - Wavelength division multiplexing (WDM).

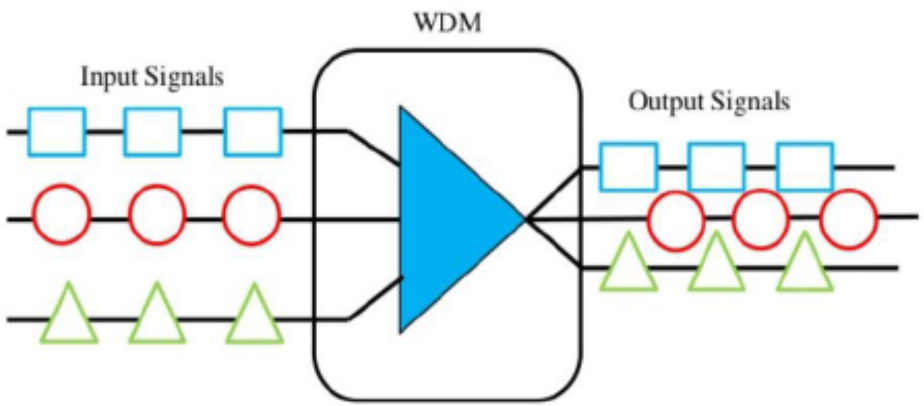
TDM vs. WDM

Complementary methods of increasing fiber capacity:

- **Time Division Multiplexing:** Increase the *serial* bitrate.
- **Wavelength Division Multiplexing:** Increase the number of *parallel* wavelengths.



- High-speed, high-bandwidth electronic equipment needed
- Expensive to build & maintain



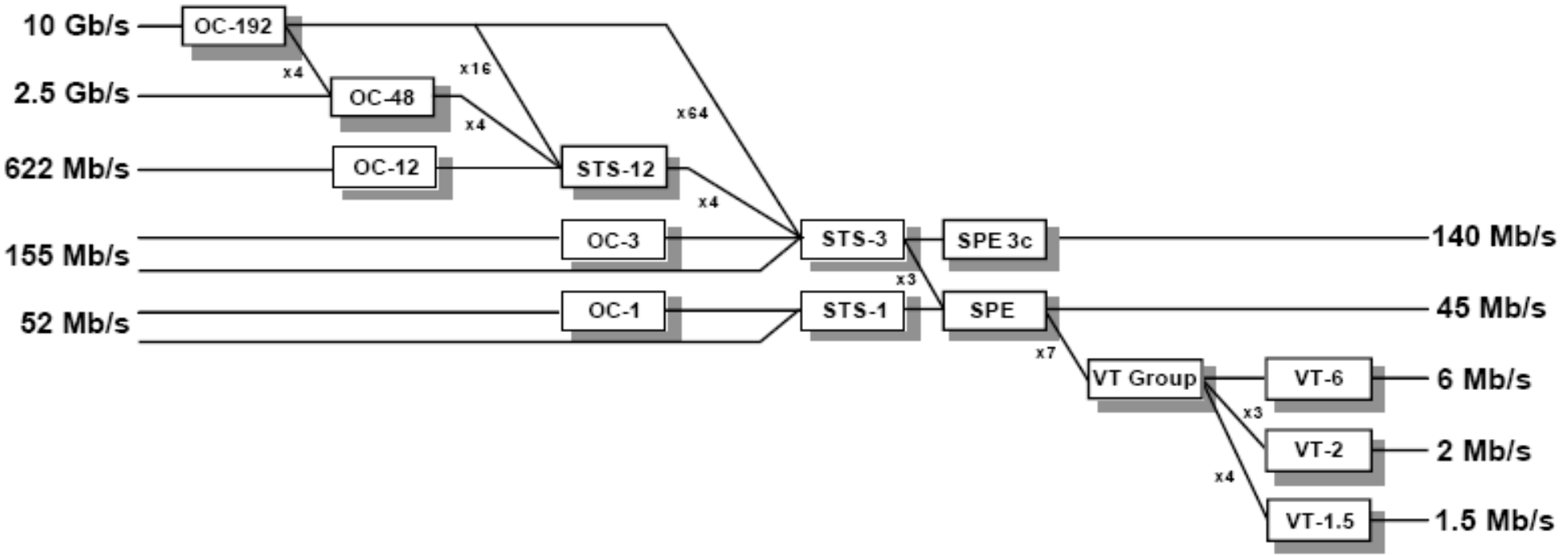
- Send data in parallel, on the **same optical fiber**
- Multiple wavelengths using lower-rate electronic/optical equipment to achieve higher throughputs overall

SONET [Optical Carrier (OC)] and SDH [Synchronous Transport Module (STM)] Levels

Optical Level	Electrical Level	Line Rate (Mbps)	Payload Rate (Mbps)	Overhead Rate (Mbps)
OC-1	STS-1	51.840	50.112	1.728
OC-3/STM-1	STS-3	155.520	150.336	5.184
OC-9/STM-3	STS-9	466.560	451.008	15.552
OC-12/STM-4	STS-12	622.080	601.344	20.736
OC-18/STM-6	STS-18	933.120	920.016	31.104
OC-24/STM-8	STS-24	1244.160	1202.688	41.472
OC-36/STM-13	STS-36	1866.240	1804.032	62.208
OC-48/STM-16	STS-48	2488.320	2405.376	82.944
OC-96/STM-32	STS-96	4976.640	4810.752	165.888
OC-192/STM-64	STS-192	9953.280	9621.504	331.776

SONET Overhead is 3% independent of Data Rate

SONET multiplexing Hierarchy



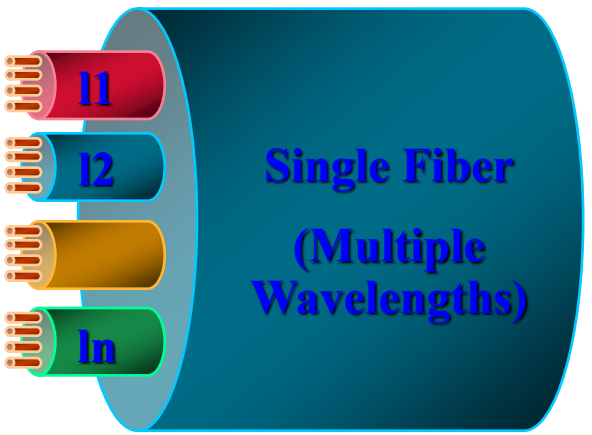
- The STS Synchronous Payload Envelope can be sub-divided into smaller components or structures known as Virtual Tributaries (VT's), for the purpose of transporting and switching payloads smaller than the STS-1 rate.
- All services below DS3 rate are transported in the VT structure. The figure illustrates the basic multiplexing structure of SONET.

TDM vs. WDM

- Time division multiplexing
  - –Single wavelength per fiber
  - –Multiple channels per fiber
  - –4 OC-3 channels in OC-12
  - –4 OC-12 channels in OC-48
  - –16 OC-3 channels in OC-48



- Wave division multiplexing
  - –Multiple wavelengths per fiber
  - –4, 16, 32, 64 wavelengths per fiber
  - –Multiple channels per wavelength



TDM vs. WDM

- TDM (SONET/SDH)
  - –Take sync and async signals and multiplex them to a single higher optical bit rate.
  - –E/O or O/E/O conversion.



- WDM
  - –Take multiple optical signals and multiplex them onto a single fiber.
  - –No signal format conversion.

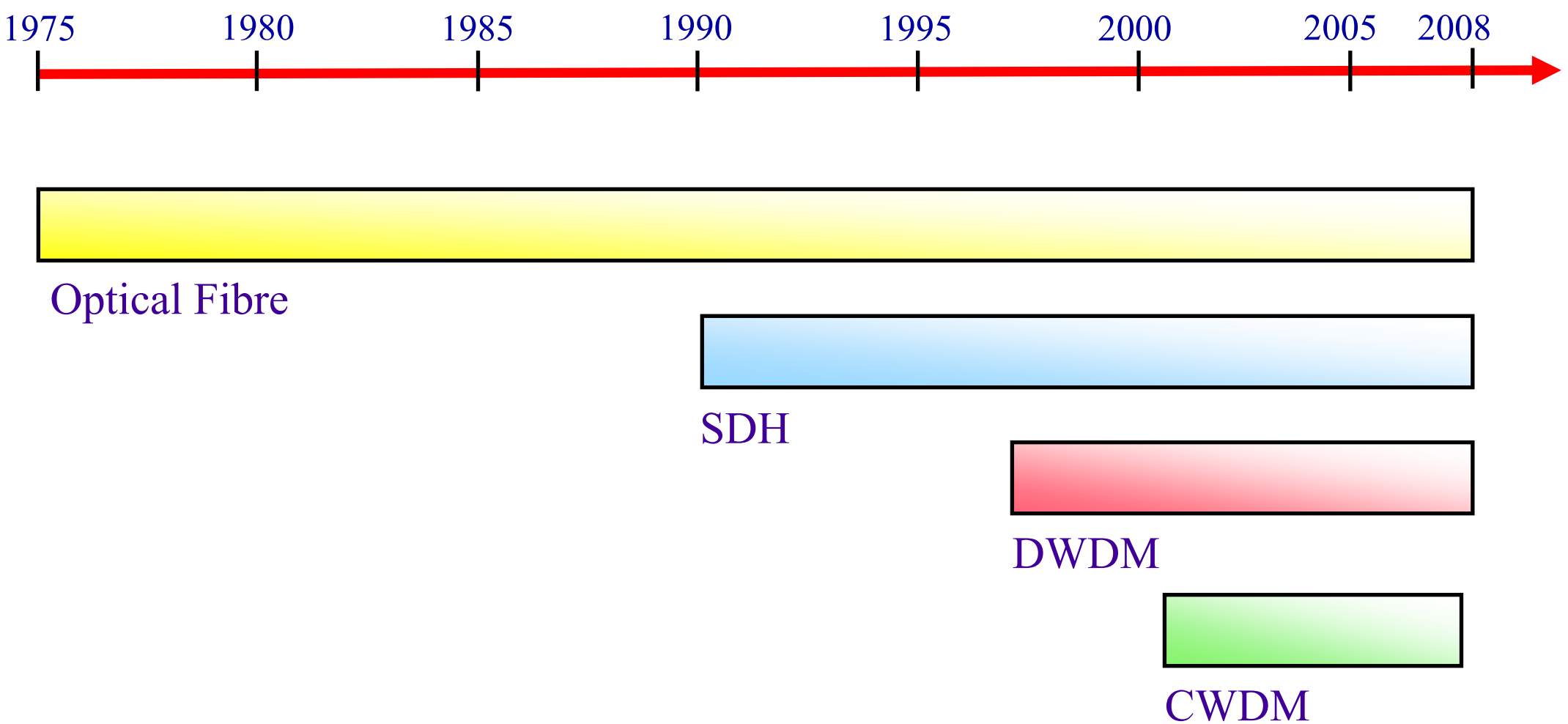


FDM vs. WDM vs. DWDM vs. UWDM

- Is WDM also a Frequency Division Multiplexing (FDM) which has been widely available for many years?
- Short Answer: Yes. There is no difference between Wavelength Division and Frequency Division. In general, FDM is used in the context of Radio Frequency (MHz–GHz) while WDM is used in the context of light (THz).
- WDM: The original standard requires 100 GHz spacing to prevent signals interference. (approximately 0.8 nm spacing at the 1550 nm wavelength)
- Dense WDM (DWDM): support multiplexing of up to 160 wavelengths of 10G/wavelength with 25GHz spacing
  - The use of sub 100 GHz for spacing is called Dense WDM.
  - Some vendors even propose to use 12.5 GHz spacing, and it would multiplex up to 320 wavelengths.

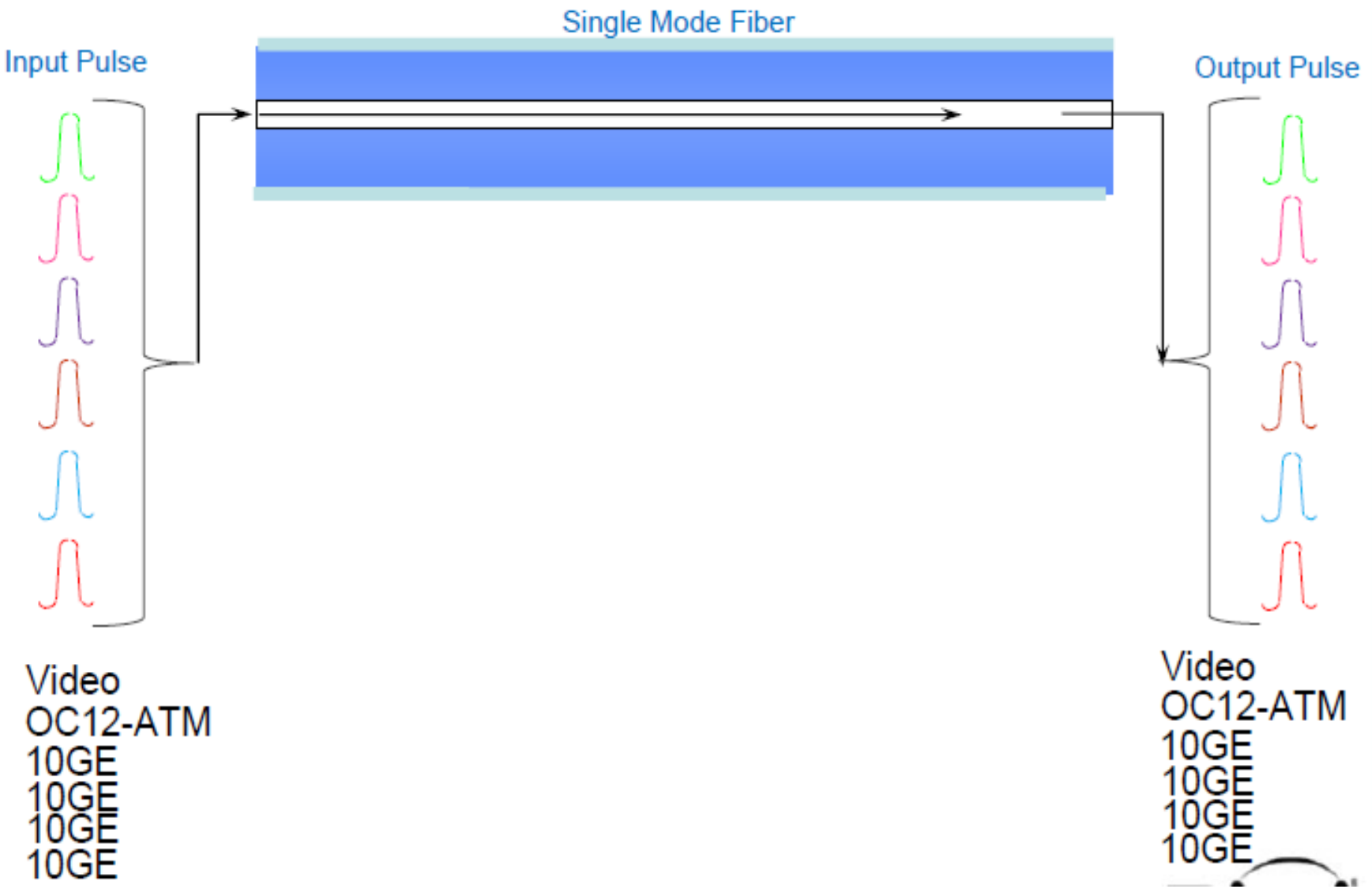


Timeline

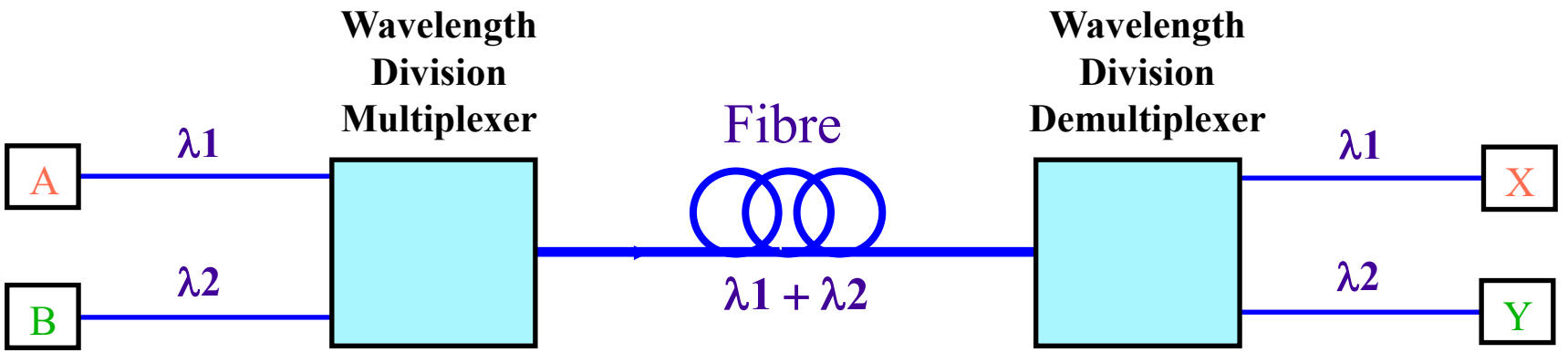

1975
1980
1985
1990
1995
2000
2005
2008
Optical Fibre
SDH
DWDM
CWDM

Wavelength Division Multiplexing (WDM)

Wavelength Division Multiplexing

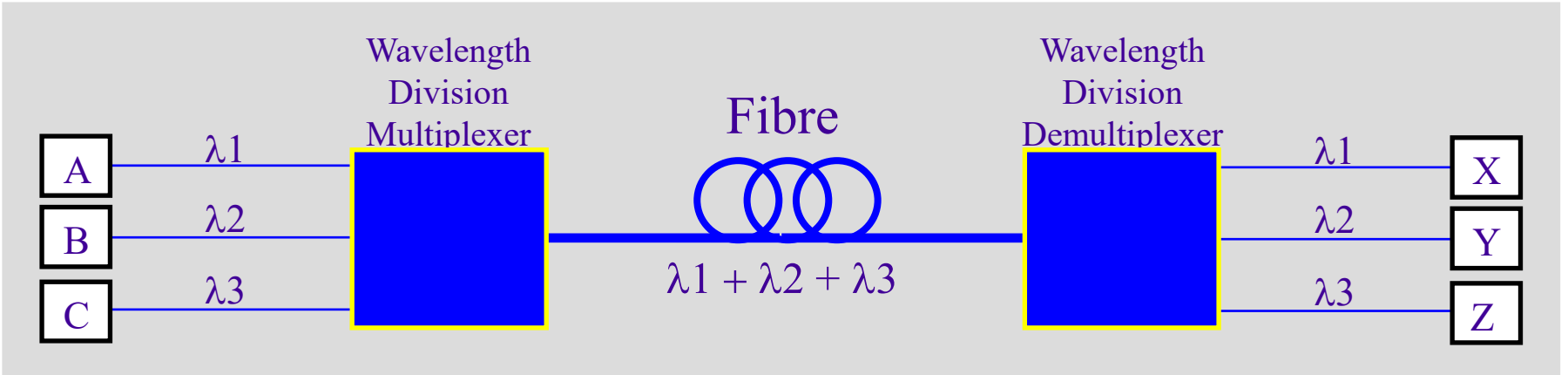


WDM Overview



- Multiple channels of information carried over the same fibre, each using an individual wavelength.
- A communicates with X and B with Y as if a dedicated fibre is used for each signal.
- Typically one channel utilises 1310 nm and the other 1550 nm.
- Broad channel spacing, several hundred nm.
- Recently WDM has become known as Coarse WDM or CWDM to distinguish it from DWDM.

WDM Overview



- Attractive multiplexing technique
  - High aggregate bit rate without high speed electronics or modulation
  - Low dispersion penalty for aggregate bit rate
  - Very useful for upgrades to installed fibres
  - Realisable using commercial components, unlike OTDM (Optical Time Division Multiplexing)
- Loss, crosstalk and non-linear effects are potential problems.

Types of WDM

	Coarse WDM (includes WWDM)	**WDM**	**DWDM** (includes ultra dense WDM)
Channel Spacings	Large, from 1.6 nm (200 GHz) to 25 nm	1310 nm lasers used in conjunction with 1550 nm lasers	Small, 200 GHz and less
Number of bands used	O,E,S,C and L	O and C	C and L
Cost per channel	Low	Low	High
Number of channels delivered	17-18 at most	2	Hundreds of channels possible
Best Application	Short-haul, Metro	PON	Long-haul

Coarse Wavelength Division Multiplexing (CWDM)

CWDM Wavelength Grid: G.694

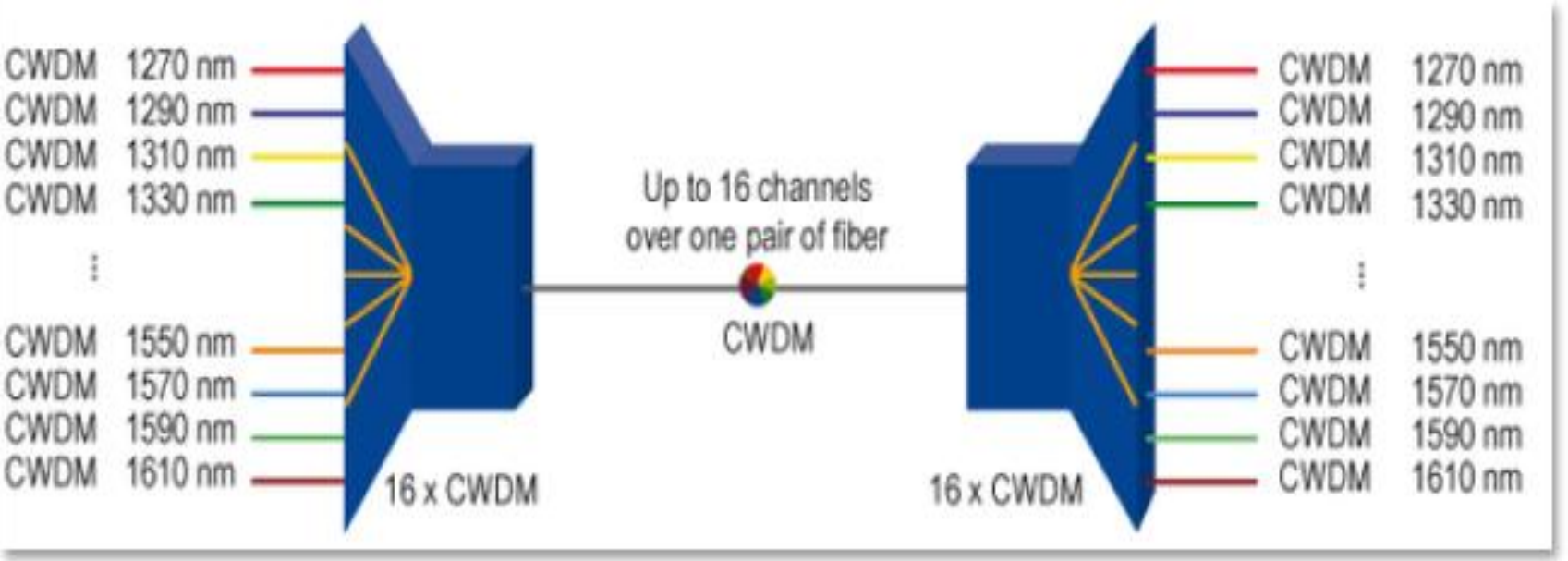
- ITU-T G.694.2 defines wavelength grids for CWDM Applications
- G.694.2 defines a wavelength grid with 20 nm channel spacing (approximately 2500 GHz)
- 18 wavelengths between 1270 nm and 1610 nm.

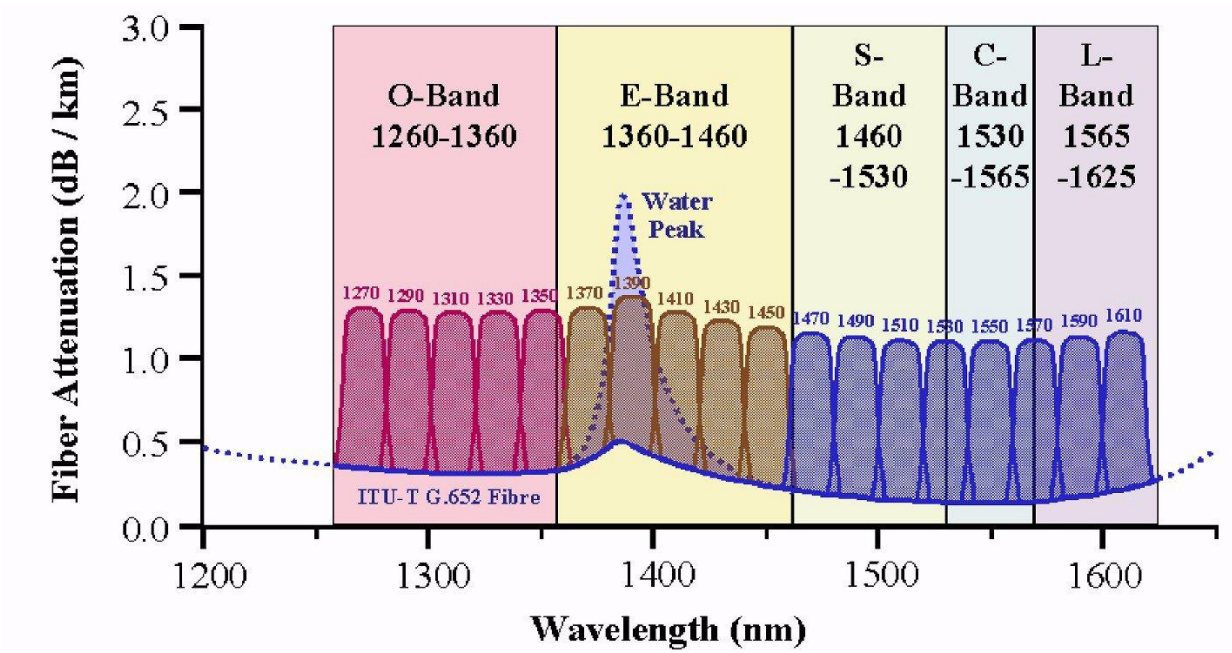
ITU CWDM Grid (nm)

1270	1290	1310	1330	1350	1370
1390	1410	1430	1450	1470	1490
1510	1530	1550	1570	1590	1610



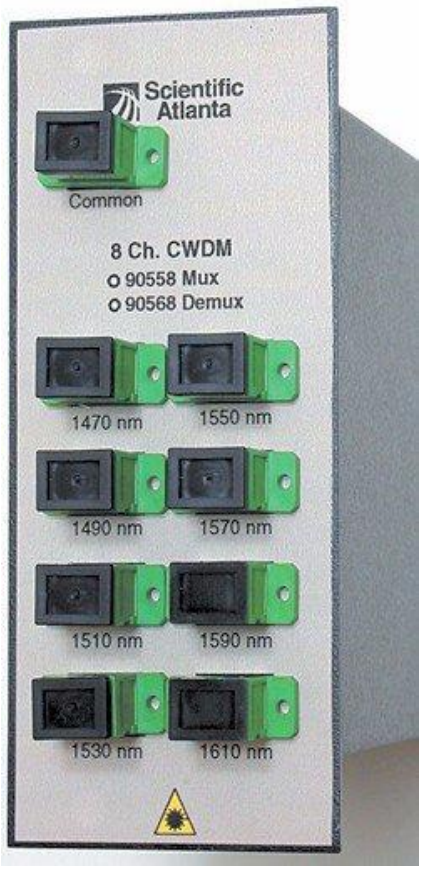
CWDM


CWDM 1270 nm
CWDM 1290 nm
CWDM 1310 nm
CWDM 1330 nm
CWDM 1550 nm
CWDM 1570 nm
CWDM 1590 nm
CWDM 1610 nm
16 x CWDM
Up to 16 channels
over one pair of fiber
CWDM
16 x CWDM
CWDM 1270 nm
CWDM 1290 nm
CWDM 1310 nm
CWDM 1330 nm
CWDM 1550 nm
CWDM 1570 nm
CWDM 1590 nm
CWDM 1610 nm


Fiber Attenuation (dB / km)
3.0
2.5
2.0
1.5
1.0
0.5
0.0
1200
1300
1400
1500
1600
Wavelength (nm)
O-Band
1260-1360
E-Band
1360-1460
S-
Band
1460
-1530
C-
Band
1530
-1565
L-
Band
1565
-1625
Water
Peak
1270 1290 1310 1330 1350
1370 1390 1410 1430 1450
1470 1490 1510 1530 1550 1570 1590 1610
ITU-T G.652 Fibre

CWDM Details

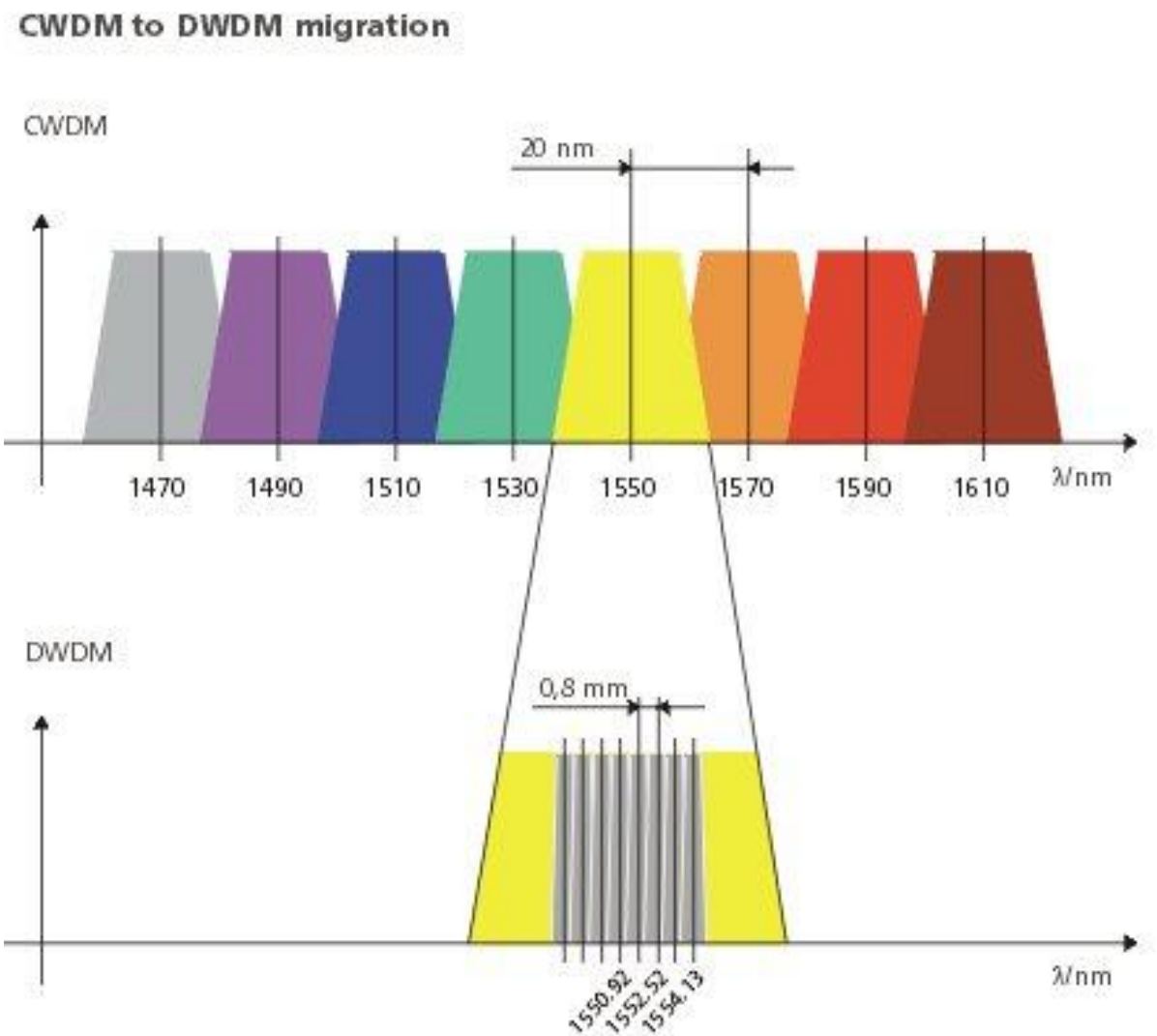
- Flexible and scalable solutions moving from 8 to 16 optical channels using two fibres for the two directions of transmission.
- Up to 8+8 optical channels using only one fibre for the two directions.
- Support for 2.5 Gbit/s provided but also support for a bit rate of 1.25 Gbit/s has been added, mainly for Gigabit-Ethernet applications.
- Two indicative link distances are covered in G.695: one for lengths up to around 40 km and a second for distances up to around 80 km.



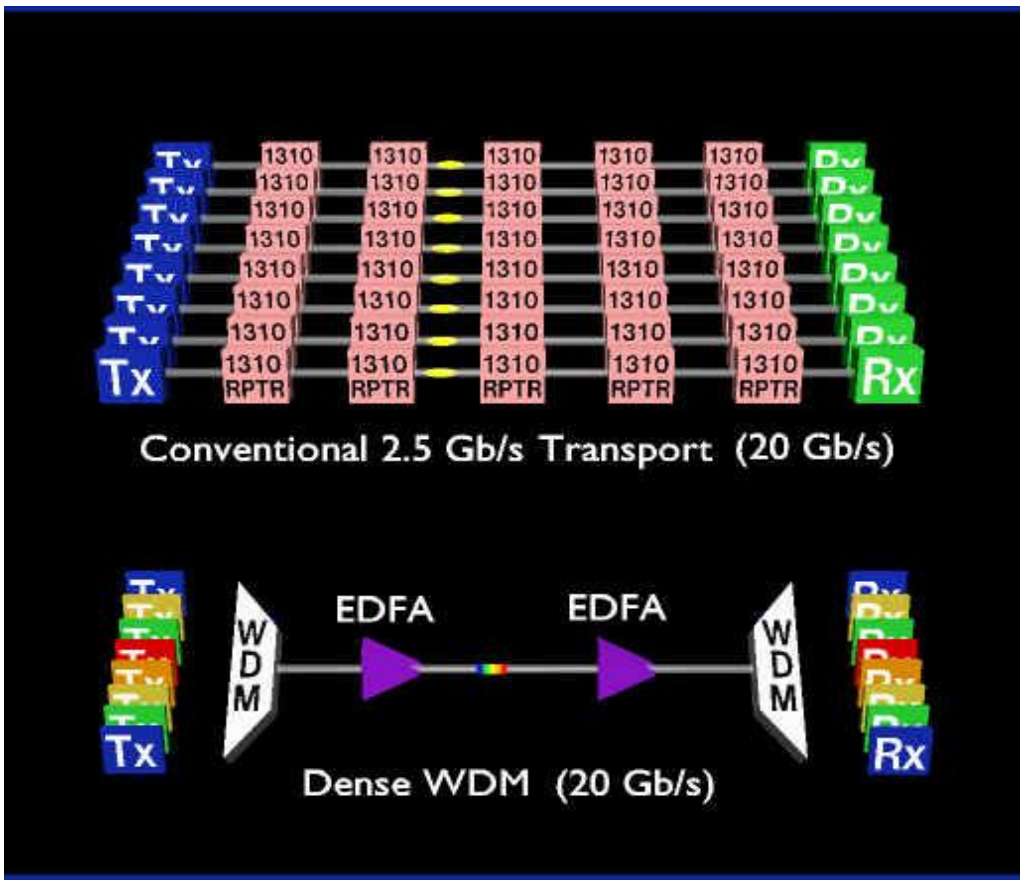
8 Ch Mux/Demux CWDM card

CWDM Migration to DWDM

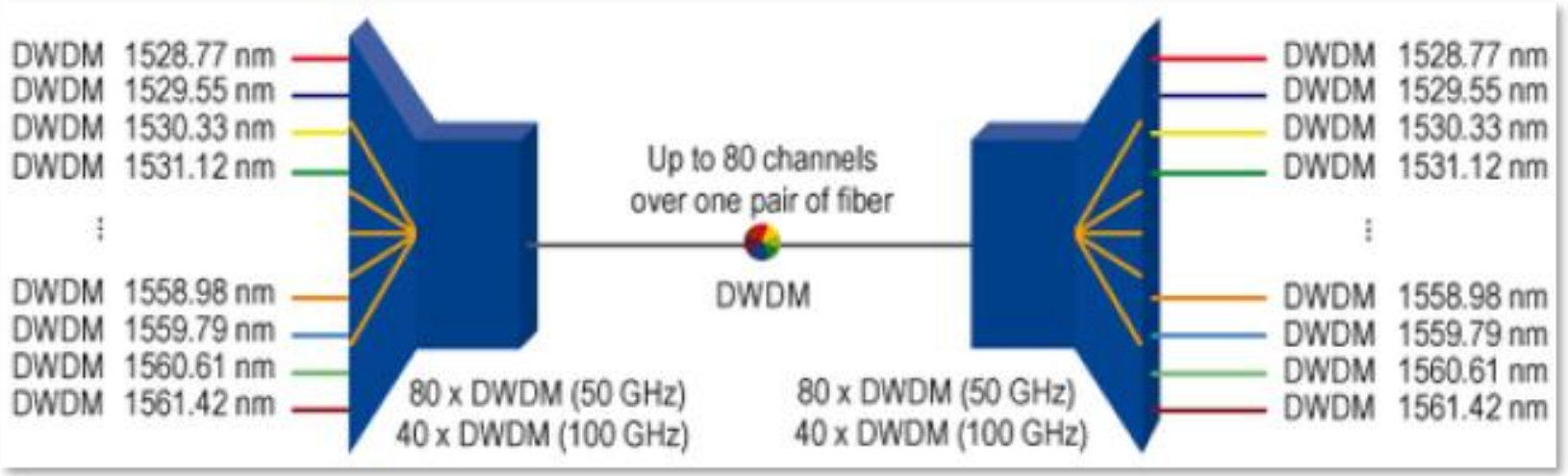
- A clear migration route from CWDM to DWDM is essential.
- Migration will occur with serious upturn in demand for bandwidth along with a reduction in DWDM costs.
- Approach involves replacing CWDM single channel space with DWDM "band".



Dense Wavelength Division Multiplexing (DWDM)



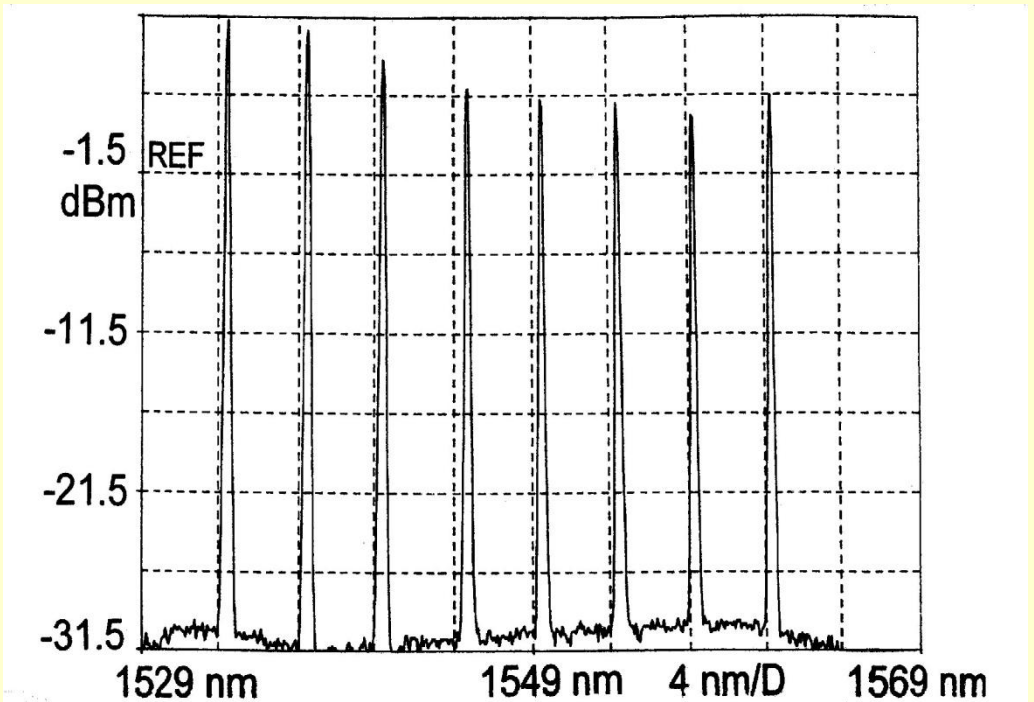
Dense Wavelength Division Multiplexing



- Multiple channels of information carried over the same fibre, each using an individual wavelength.
- *Dense WDM* is WDM utilizing closely spaced channels.
- ITU 694.1 defines channel spacing with 1.6 nm, 0.8 nm and 0.4 nm.
- Commercial systems available with capacities of 32 channels and upwards; > 80 Gb/s per fibre.

Early DWDM: CNET 160 Gbits/sec WDM

- 160 Gbits/s
- 8 channels, 20 Gbits/s each
- Grating multiplex/demultiplex
- 4 nm channel spacing
- 1533 to 1561 nm band
- 238 km span
- 3 optical amplifiers used



Multiplexer Optical Output Spectrum for an 8 DWDM channel system, showing individual channels

Sample DWDM Signal

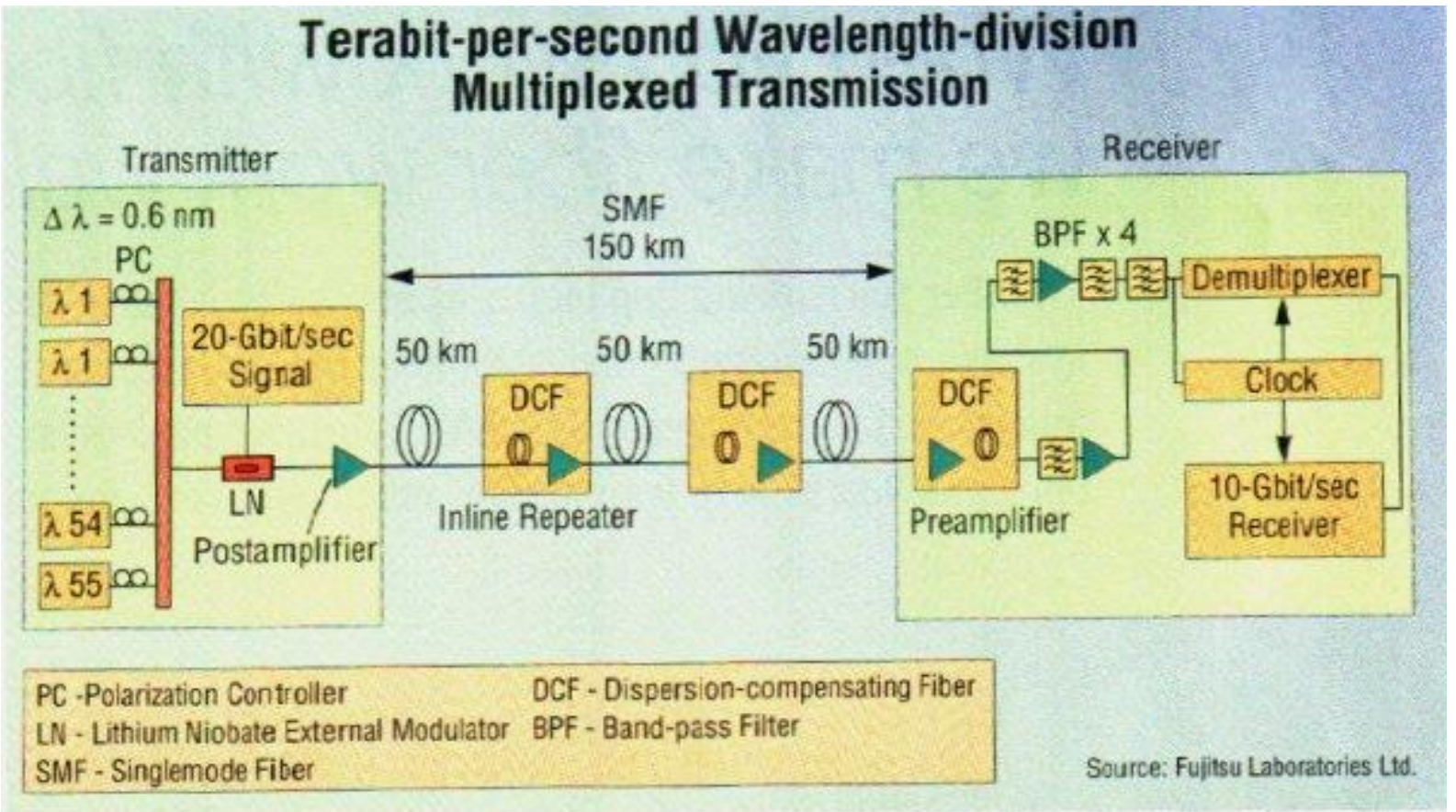
Art O'Hare, CNET, PTL May 1996

DWDM: Key Issues

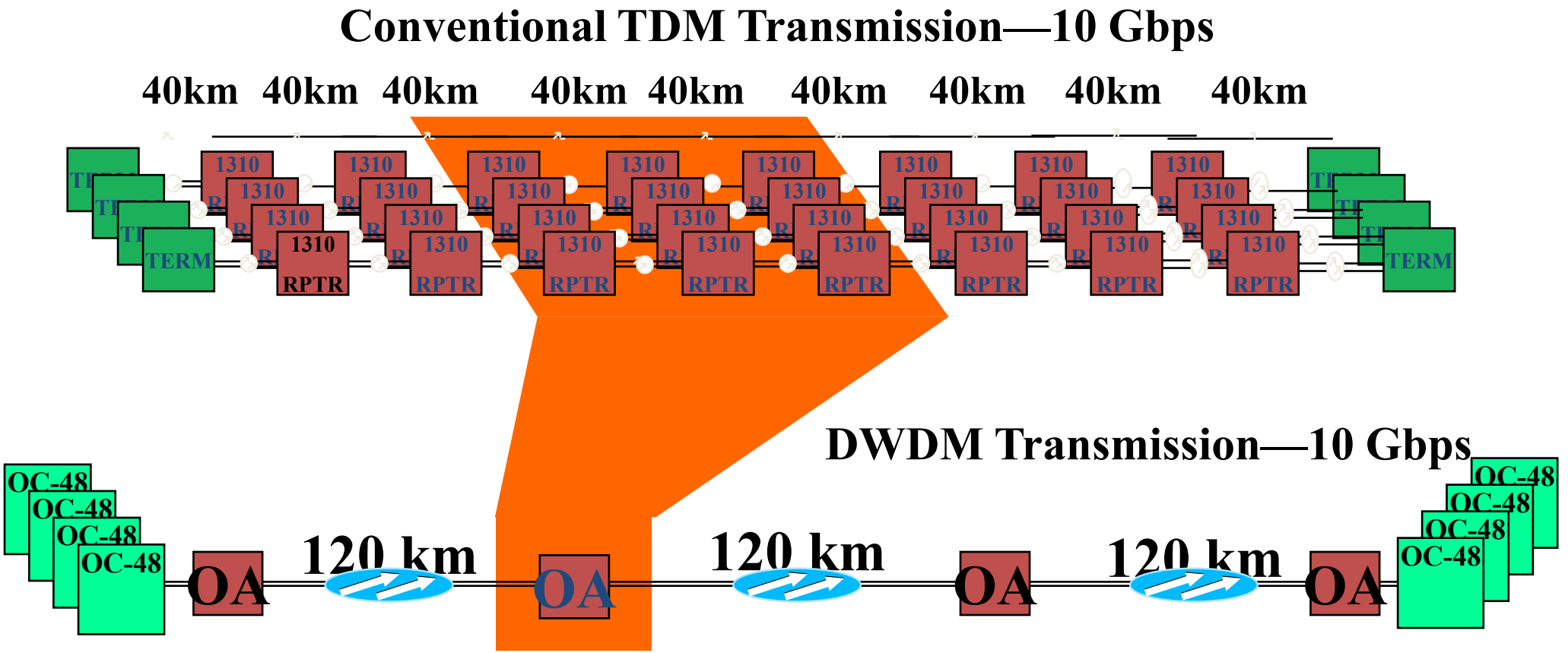
- *Dense WDM* is WDM utilizing closely spaced channels.
- Channel spacing reduced to 1.6 nm and less.
- Cost effective way of increasing capacity without replacing fibre.
- Commercial systems available with capacities of 32 channels and upwards; > 80 Gb/s per fibre.
- Allows new optical network topologies, for example high speed metropolitian rings.
- Optical amplifiers are also a key component.

Terabit Transmission using DWDM

- 1.1 Tbits/sec total bit rate (more than 13 million telephone channels)
- 55 wavelengths at 20 Gbits/sec each
- 1550 nm operation over 150 km with dispersion compensation
- Bandwidth from 1531.7 nm to 1564.07 nm (0.6 nm spacing)



DWDM Economy



4 Fiber Pairs

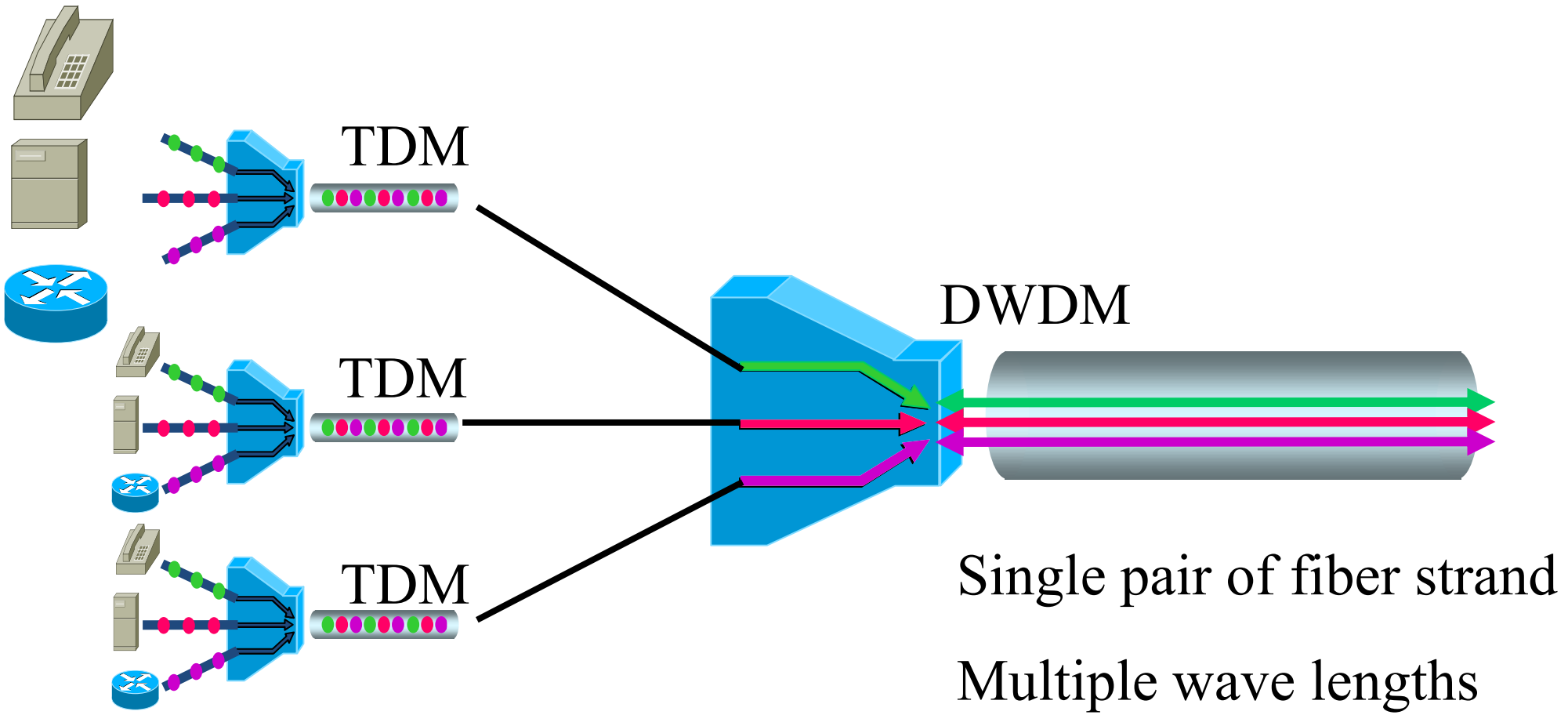
32 Regenerators

1 Fiber Pair

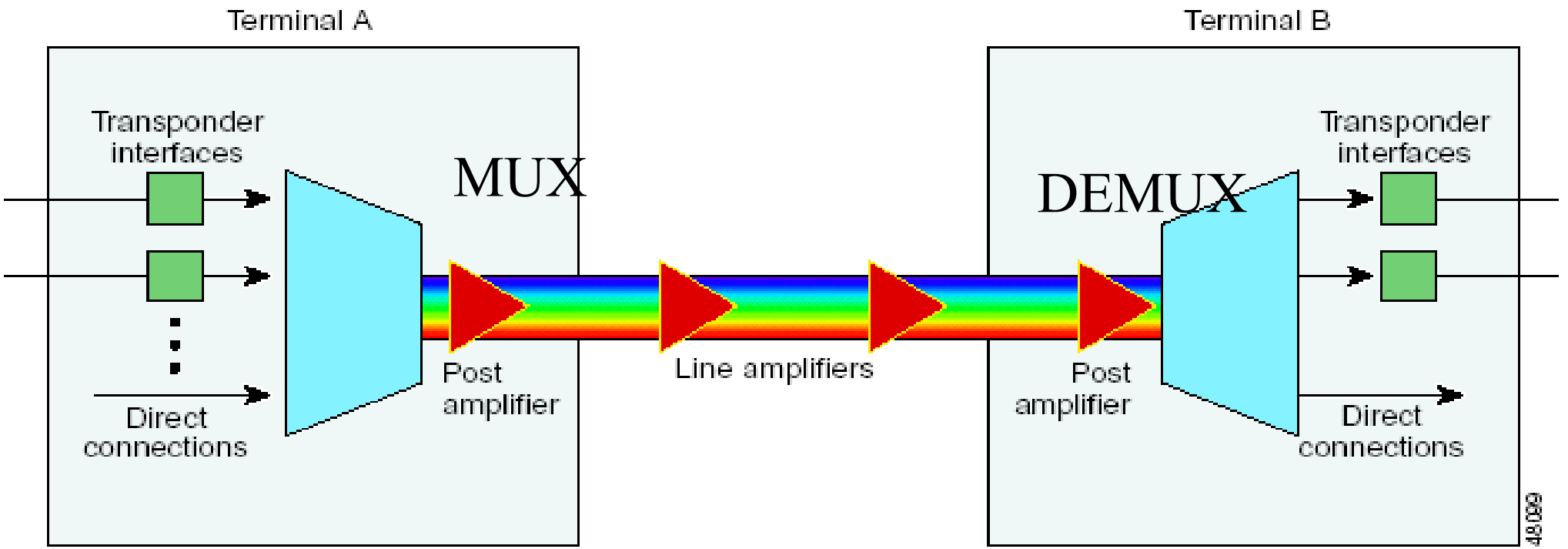
4 Optical Amplifiers

DWDM: How does it work?

TDM: multiple services onto a single wavelength



DWDM Network



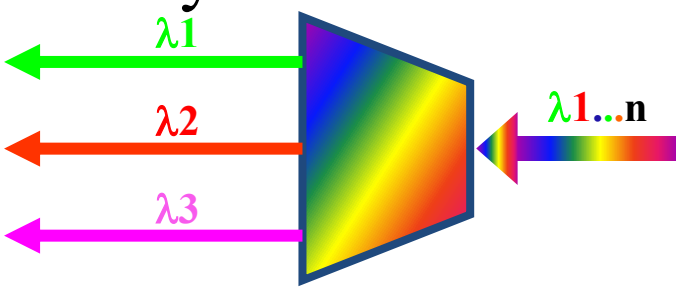
DWDM Network Components



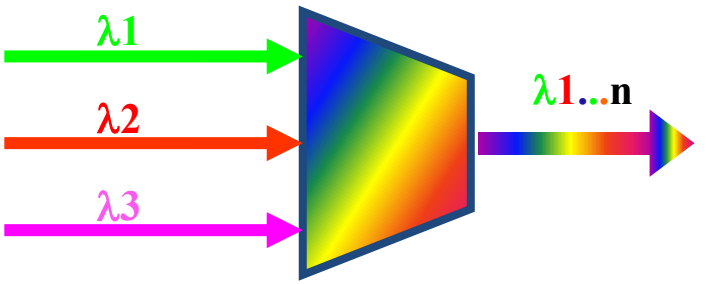
Transponder

Optical λ =< DWDM λ

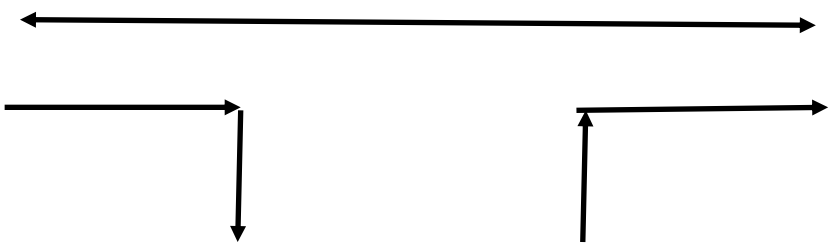
Usually do O-E-O



Optical De-multiplexer



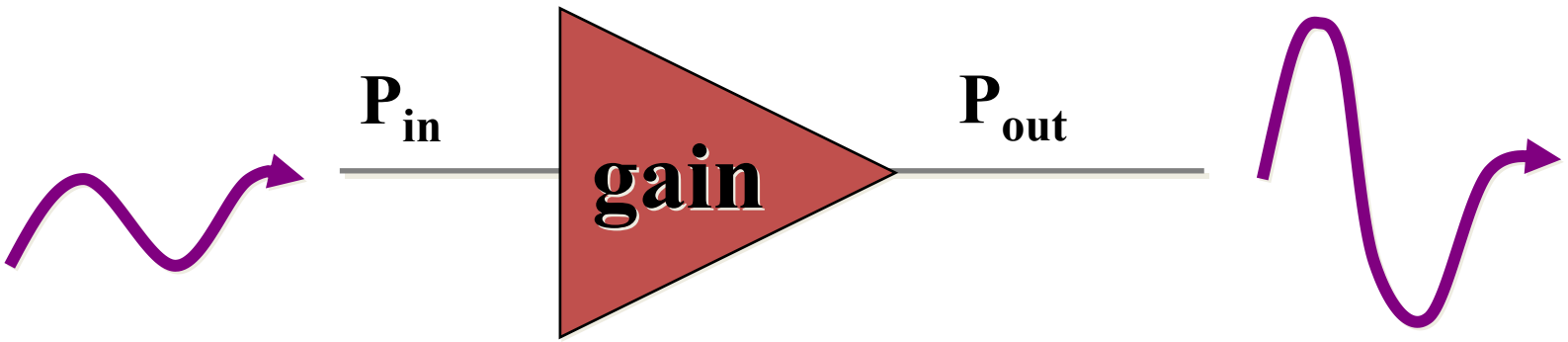
Optical Multiplexer



Optical Add/Drop Multiplexer

(OADM)

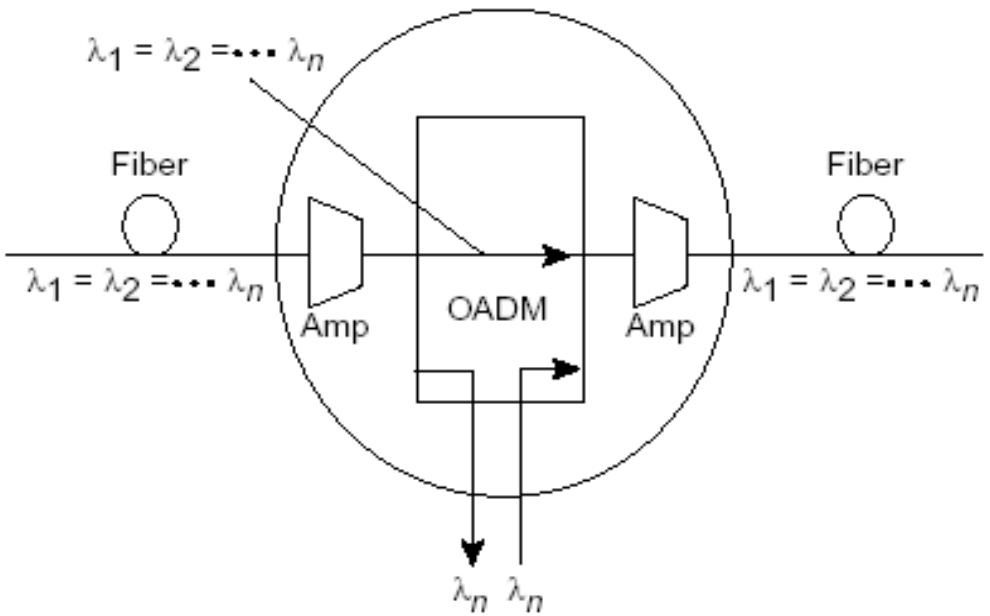
Optical Amplifier (OA)



- EDFA (Erbium Doped Fiber Amplifier) amplifier
- Separate amplifiers for C-band and L-band

Optical ADM (OADM)

- OADM is similar in many respects to SONET ADM, except that only optical wavelengths are added and dropped, and there is no conversion of the signal from optical to electrical.



Q: there is no framing of DWDM, so how do we add/drop/pass *light*?

A: λ It is based on λ and λ only.

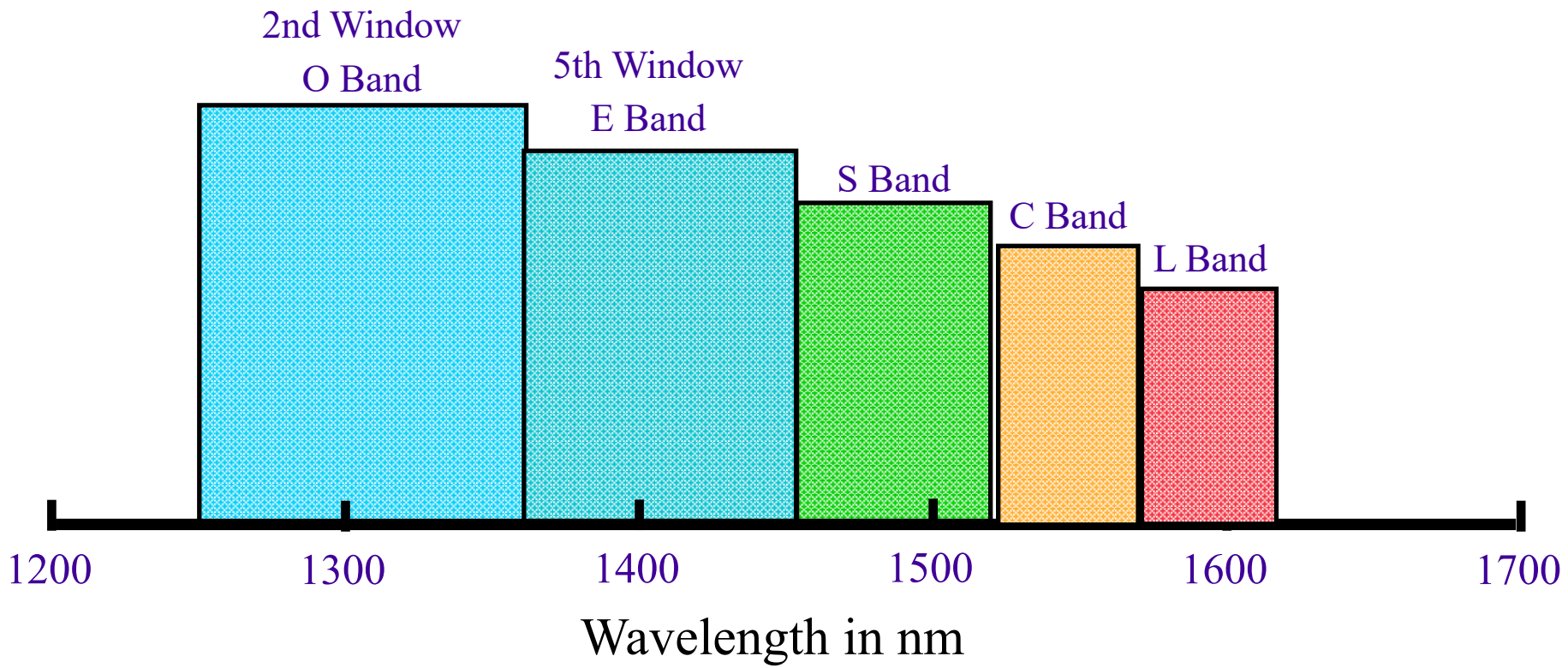
DWDM Standards



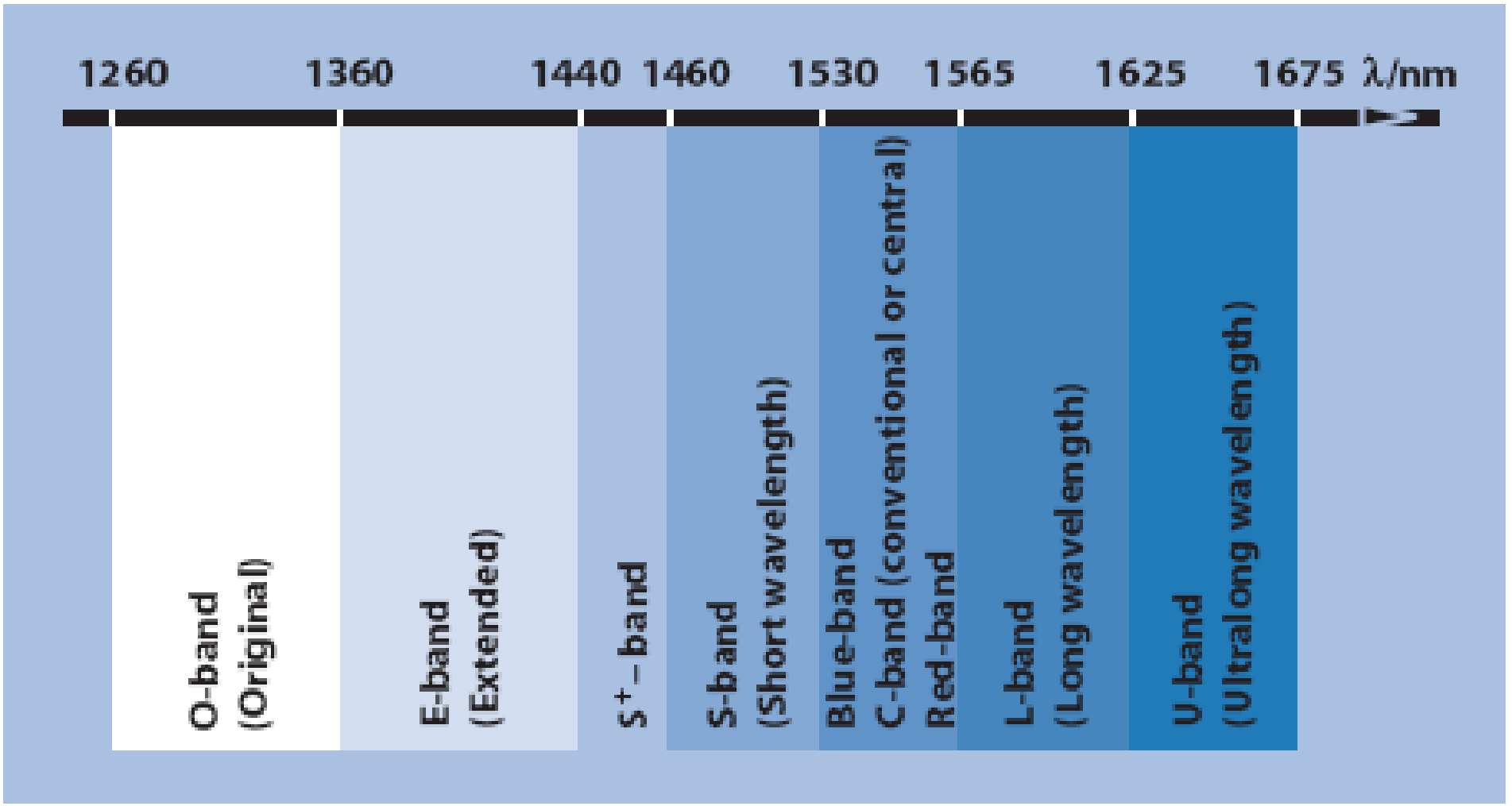
DWDM Standards

- ITU Recommendation is G.692 "Optical interfaces for multichannel systems with optical amplifiers“.
- G.692 includes a number of DWDM channel plans.
- Channel separation set at:
  - 50, 100 and 200 GHz
  - equivalent to approximate wavelength spacing of 0.4, 0.8 and 1.6 nm
- Channels lie in the range 1530.3 nm to 1567.1 nm (so-called C-Band).
- Newer "L-Band" exists from about 1570 nm to 1620 nm.
- Supervisory channel also specified at 1510 nm to handle alarms and monitoring.

Optical Spectral Bands



Optical Spectral Bands



Channel Spacing

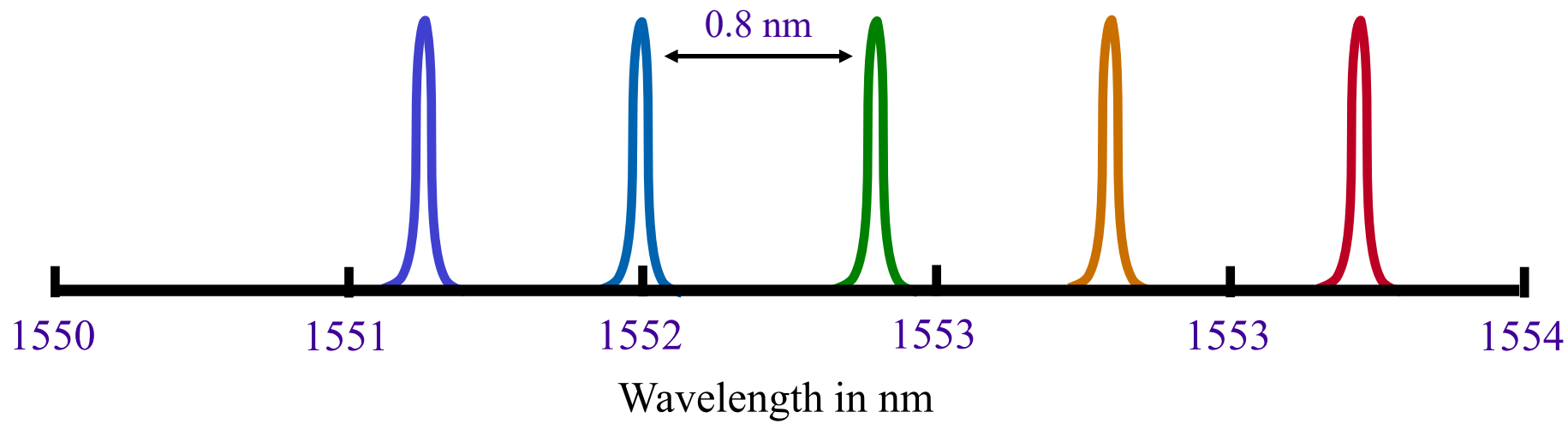
- Trend is toward smaller channel spacing, to increase the channel count.

Dense Wave Division Multiplexing (DWDM)

- ITU channel spacing are 0.4 nm, 0.8 nm and 1.6 nm (50, 100 and 200 GHz).

Ultra Dense Wave Division Multiplexing (U-DWDM)

- ITU channel spacing of 0.2 nm (25 GHz) and even 0.1 nm (12.5 GHz).
- Requires laser sources with excellent long term wavelength stability, better than 10 pm.
- One target is to allow more channels in the C-band without other upgrades.



ITU DWDM Channel Plan
0.4 nm Spacing (50 GHz)

So called
ITU *C-Band*

81 channels defined

Another band called the
L-band exists
above 1565 nm

All Wavelengths in nm

1528.77	1534.64	1540.56	1546.52	1552.52	1558.58
1529.16	1535.04	1540.95	1546.92	1552.93	1558.98
1529.55	1535.43	1541.35	1547.32	1553.33	1559.39
1529.94	1535.82	1541.75	1547.72	1553.73	1559.79
1530.33	1536.22	1542.14	1548.11	1554.13	1560.20
1530.72	1536.61	1542.54	1548.51	1554.54	1560.61
1531.12	1537.00	1542.94	1548.91	1554.94	
1531.51	1537.40	1543.33	1549.32	1555.34	
1531.90	1537.79	1543.73	1549.72	1555.75	
1532.29	1538.19	1544.13	1550.12	1556.15	
1532.68	1538.58	1544.53	1550.52	1556.55	
1533.07	1538.98	1544.92	1550.92	1556.96	
1533.47	1539.37	1545.32	1551.32	1557.36	
1533.86	1539.77	1545.72	1551.72	1557.77	
1534.25	1540.16	1546.12	1552.12	1558.17	

Speed of Light assumed to be 2.99792458 x 10^8 m/s

ITU DWDM Channel Plan
0.8 nm Spacing (100 GHz)

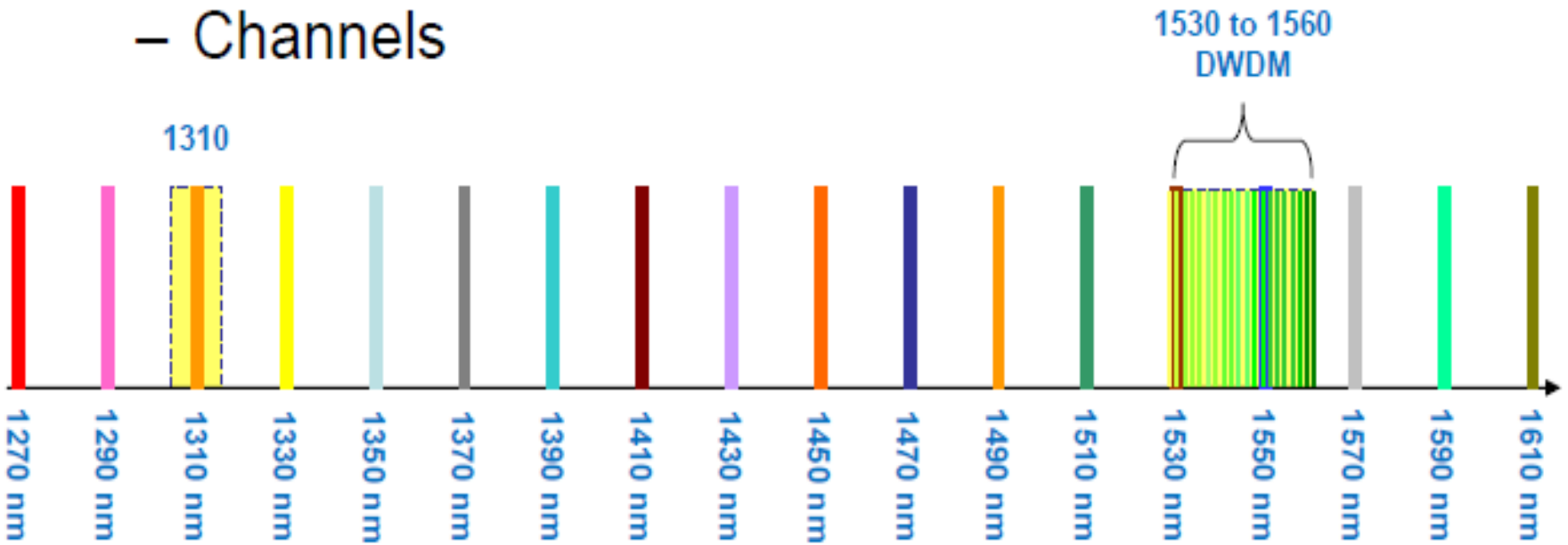
All Wavelengths in nm

1528.77	1534.64	1540.56	1546.52	1552.52	1558.98
1529.55	1535.43	1541.35	1547.32	1553.33	1559.79
1530.33	1536.22	1542.14	1548.11	1554.13	1560.61
1531.12	1537.00	1542.94	1548.91	1554.94	
1531.90	1537.79	1543.73	1549.72	1555.75	
1532.68	1538.58	1544.53	1550.52	1556.55	
1533.47	1539.37	1545.32	1551.32	1557.36	
1534.25	1540.16	1546.12	1552.12	1558.17	

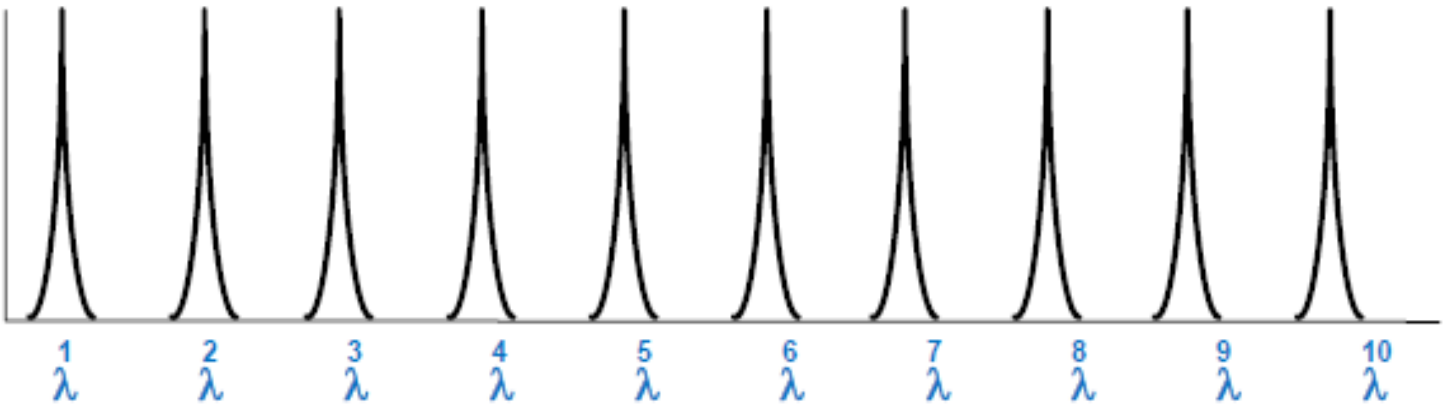
Speed of Light assumed to be 2.99792458 x 10^8 m/s

DWDM

- Dense Wave Division Multiplexing
- Standard channel plan developed by the ITU
  - International Telecommunications Union
  - 400, 200, 100, and now 50 GHz spacing between channels
  - Starting at 1530nm and going thru 1560nm
  - Channels

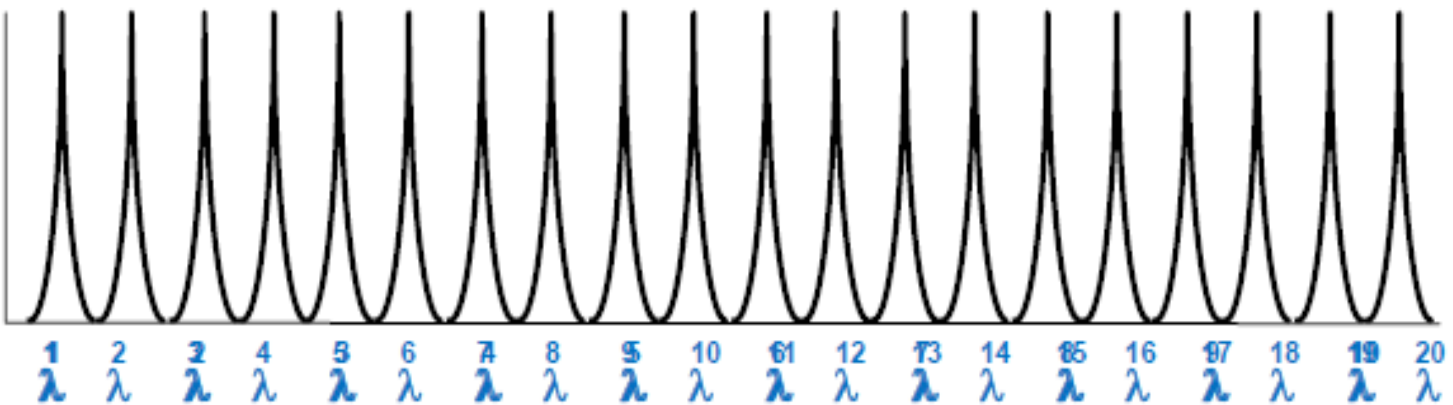


Channel Spacing (400 GHz and 200 GHz)



400 GHz Spacing

10 Channels in Amplified C Band

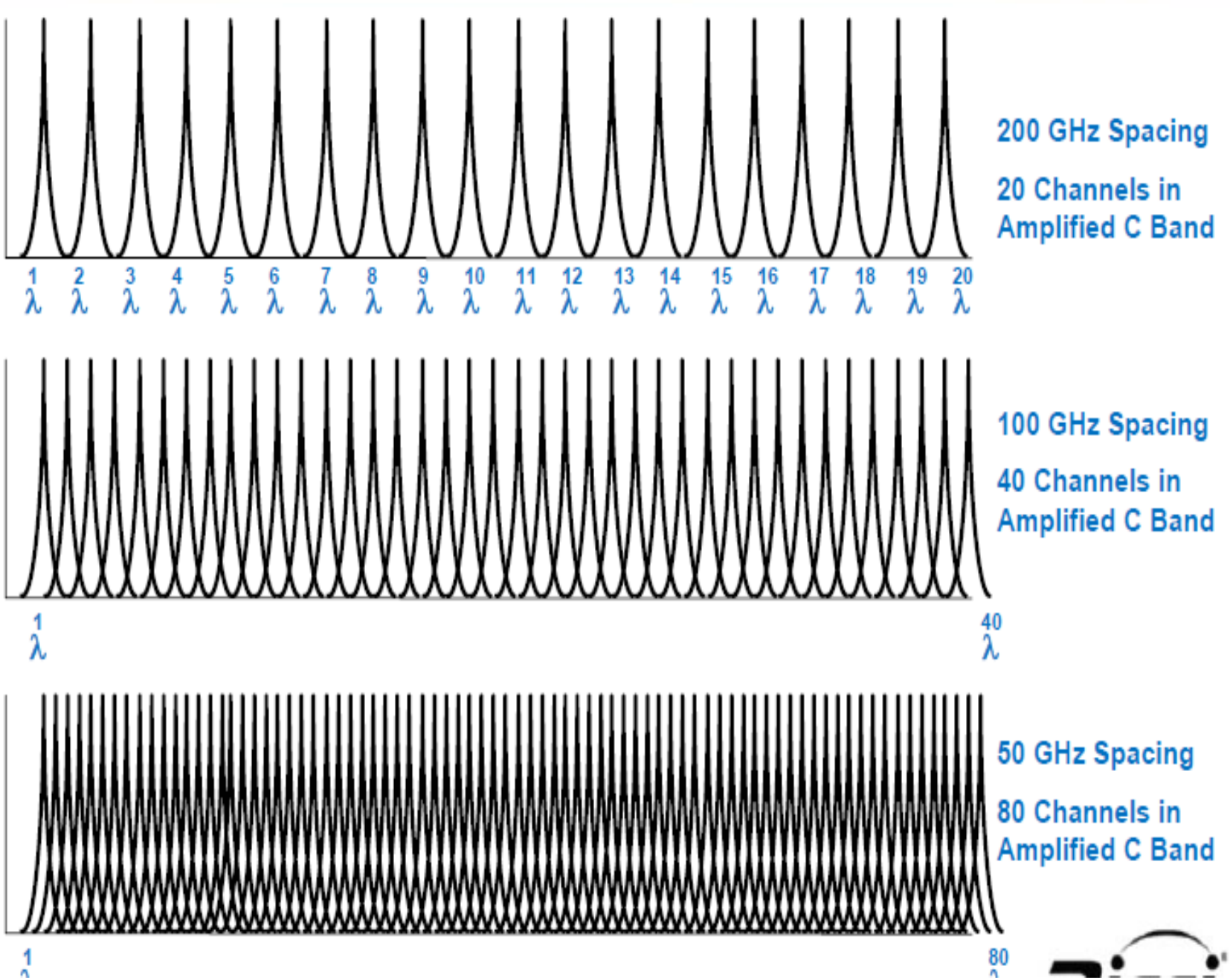


200 GHz Spacing

20 Channels in Amplified C Band

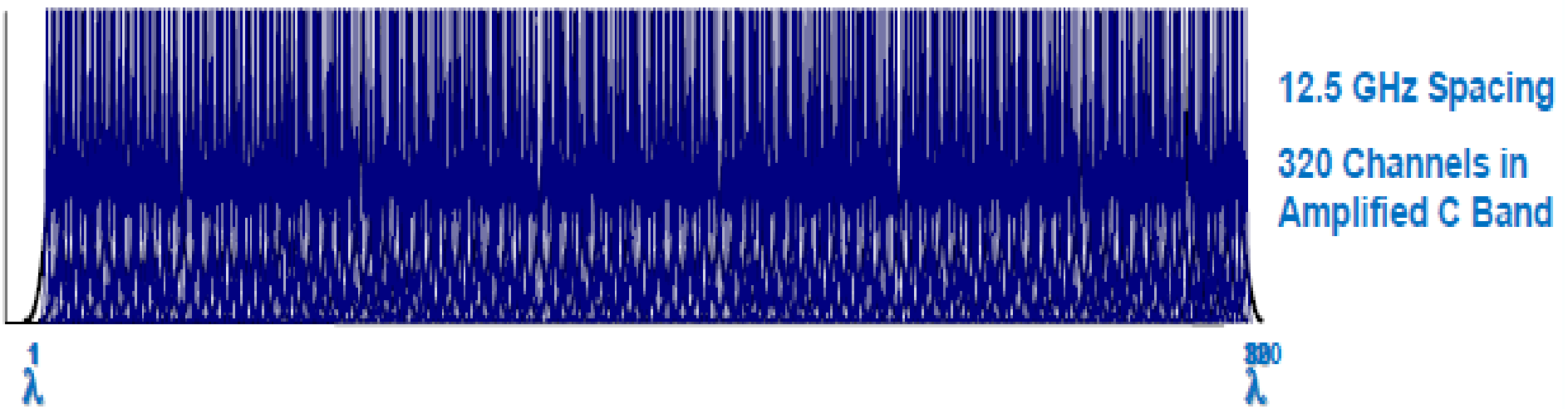
- Spacing between carriers is reduced by half

Channel Spacing (100 GHz and 50 GHz)



U-DWDM

- Ultra Dense Wave Division Multiplexing (U-DWDM)
- Standard channel plan developed by the ITU
  - International Telecommunications Union
  - 12.5 GHz spacing between channels
  - Starting at 1530nm and going thru 1560nm
  - 320 Channels



G.692 Representation of a Standard DWDM System

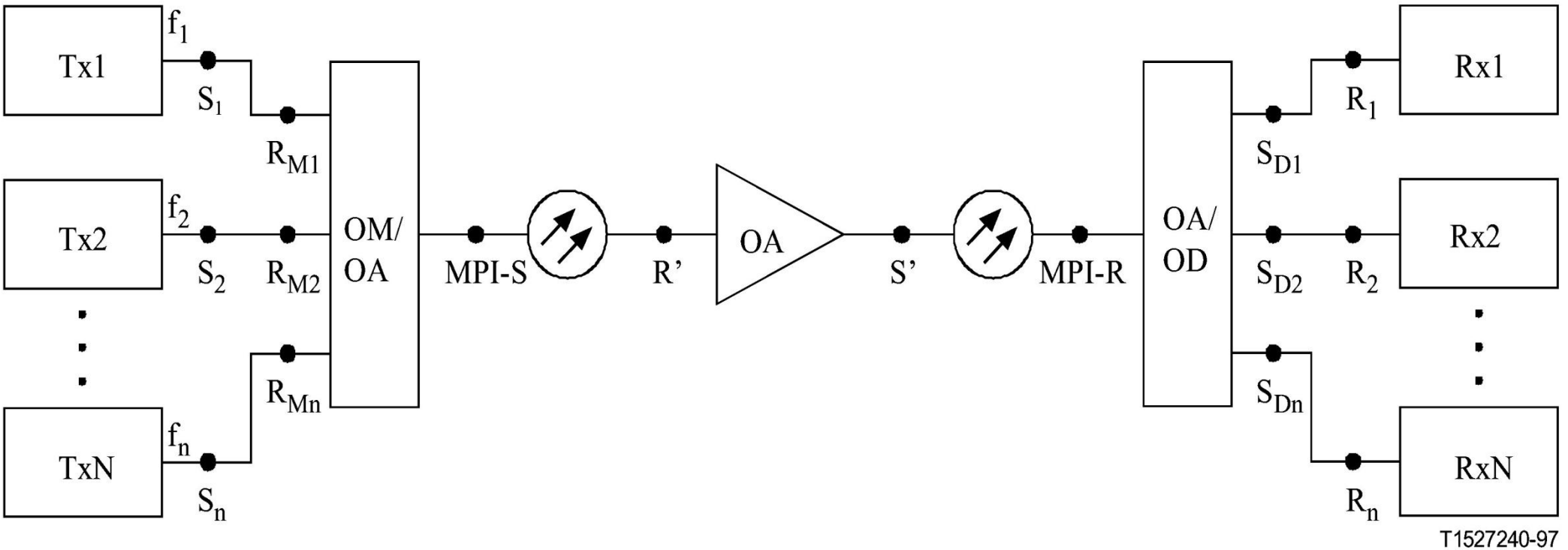
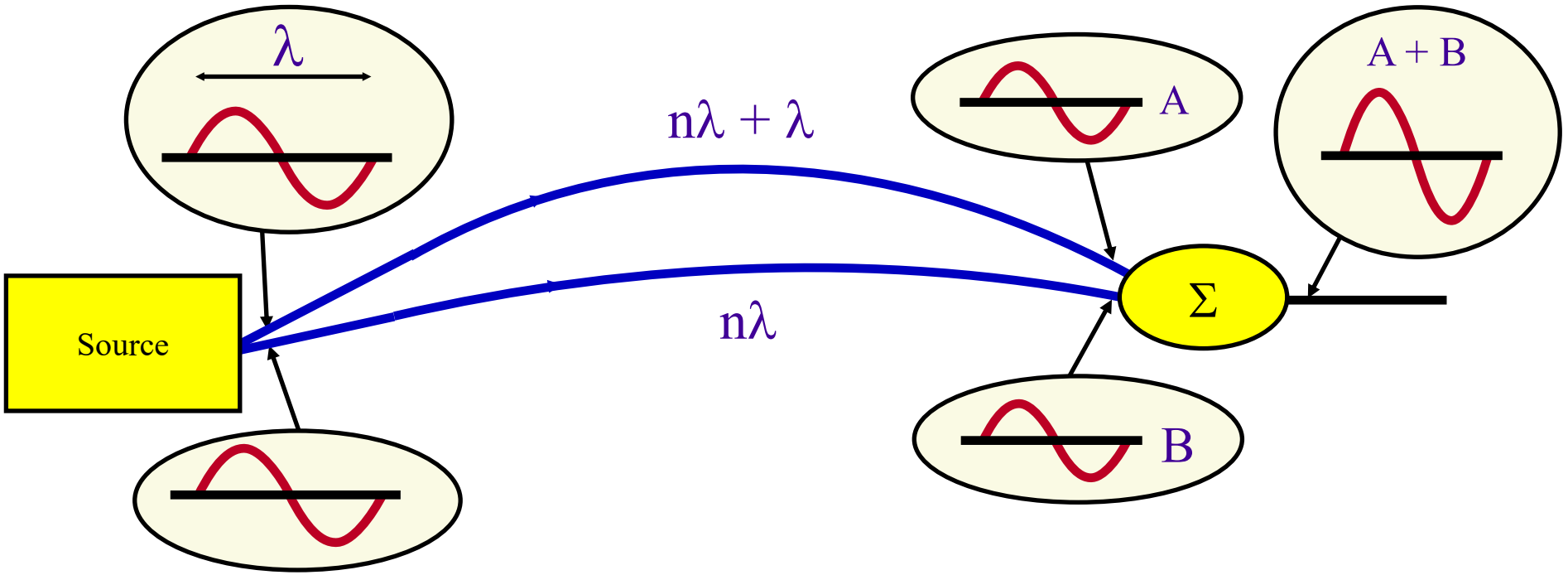


Figure 1/G.692 – Representation of optical line system interfaces

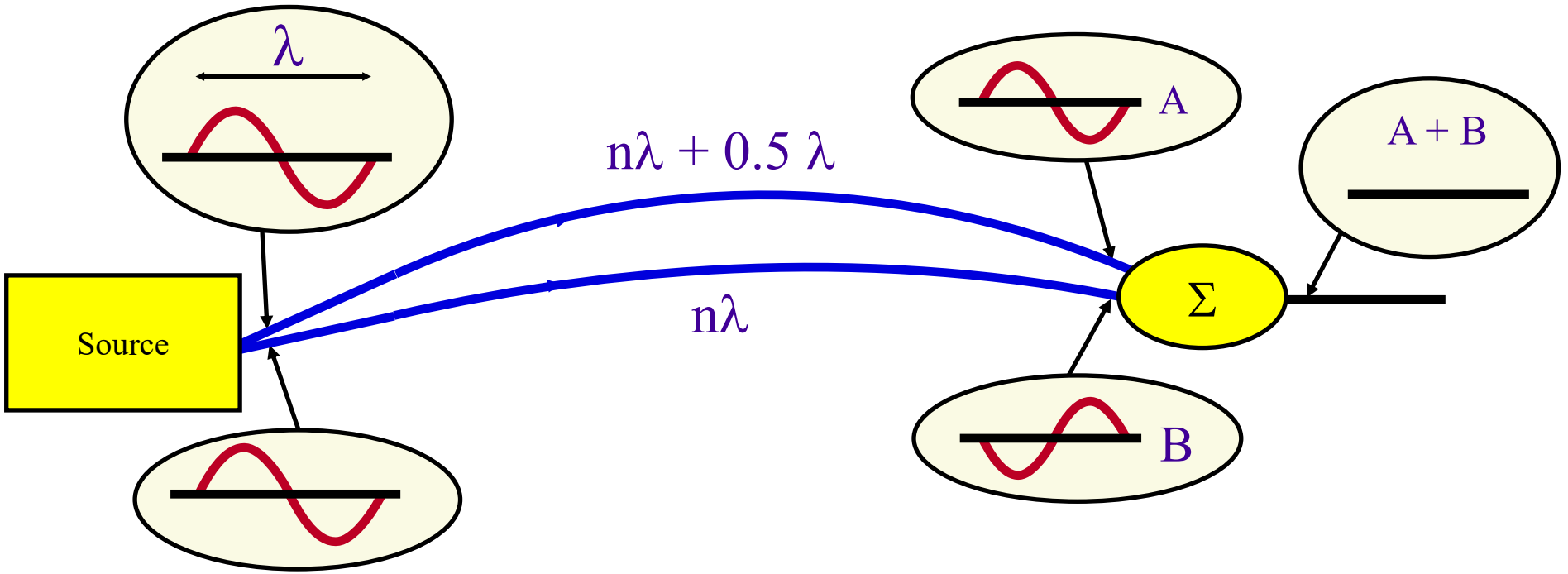
Mux/Demuxes

Constructive Interference



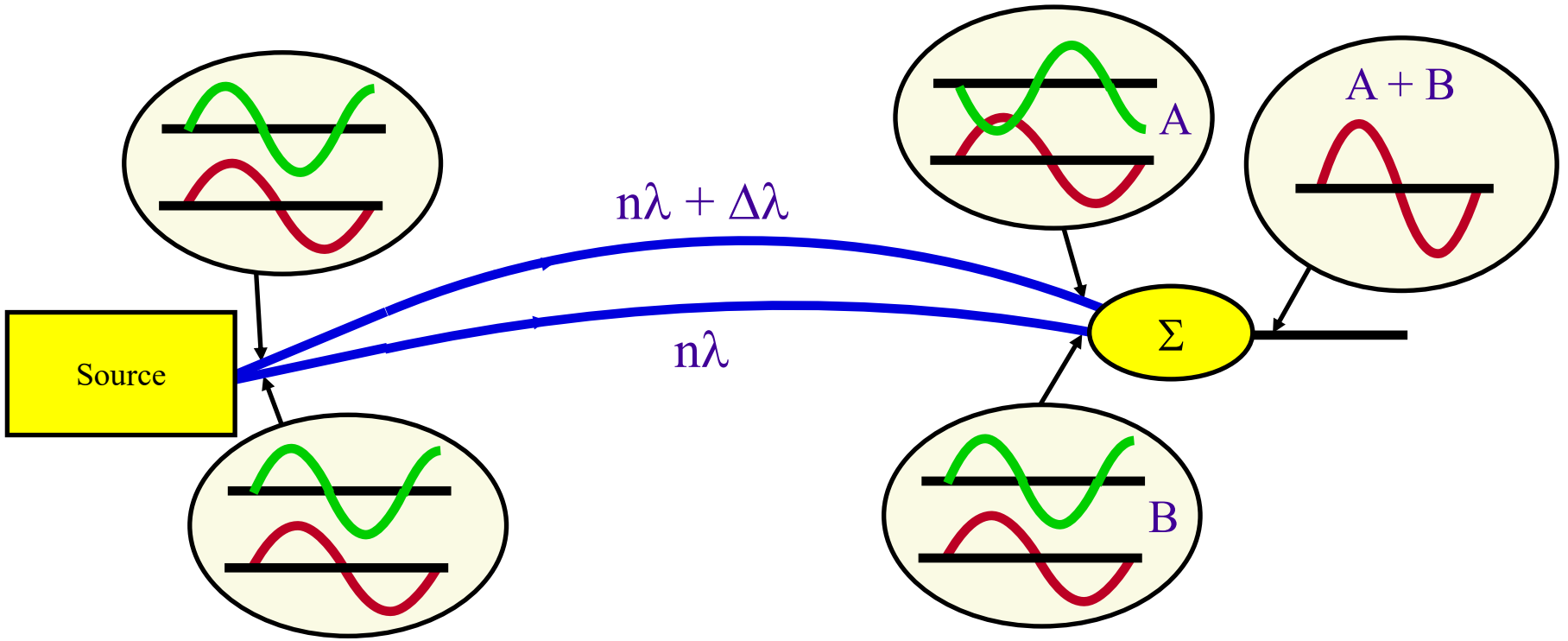
- Travelling on two different paths, both waves recombine (at the summer, Σ).
- Because of the λ path length difference the waves are in-phase.
- Complete reinforcement occurs, so-called constructive interference.

Destructive Interference



- Travelling on two different paths, both waves recombine (at the summer, Σ).
- Because of the 0.5λ path length difference the waves are out of phase.
- Complete cancellation occurs, so-called destructive interference.

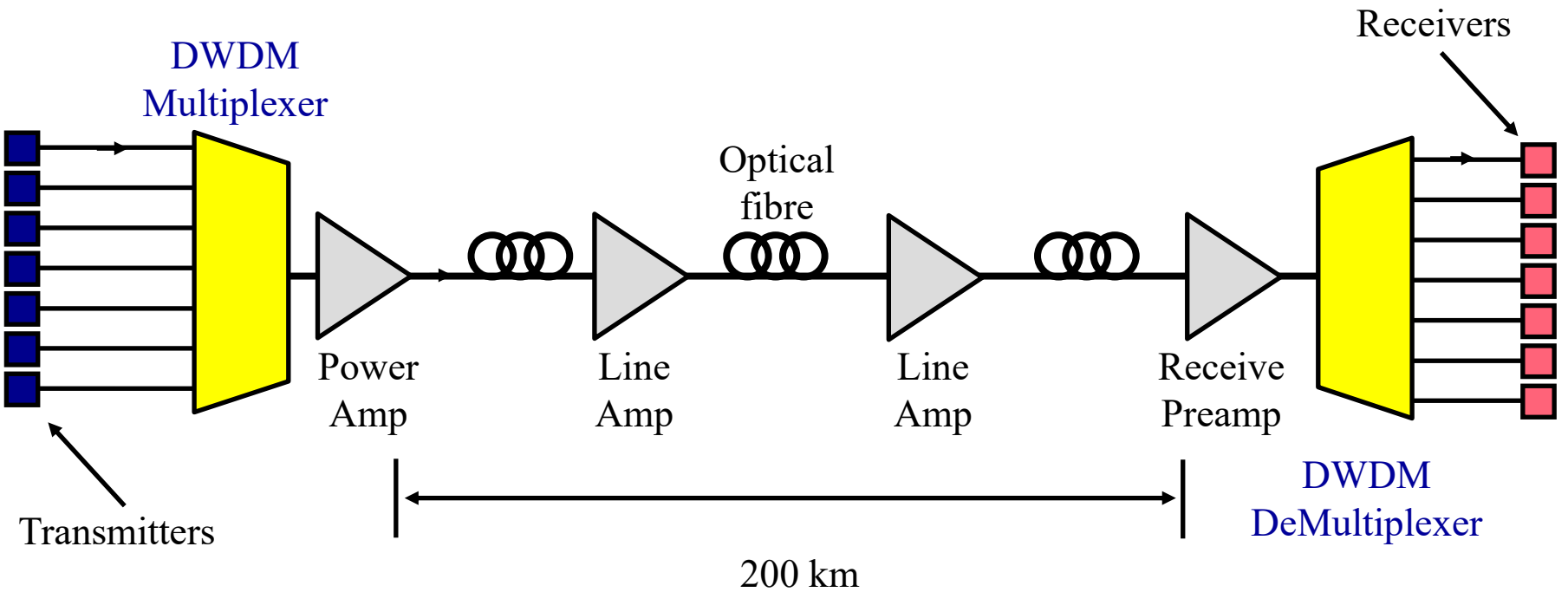
Using Interference to Select a Wavelength



- Two different wavelengths, both travelling on two different paths.
- Because of the path length difference the "Red" wavelength undergoes constructive interference while the "Green" suffers destructive interference.
- Only the Red wavelength is selected, Green is rejected.

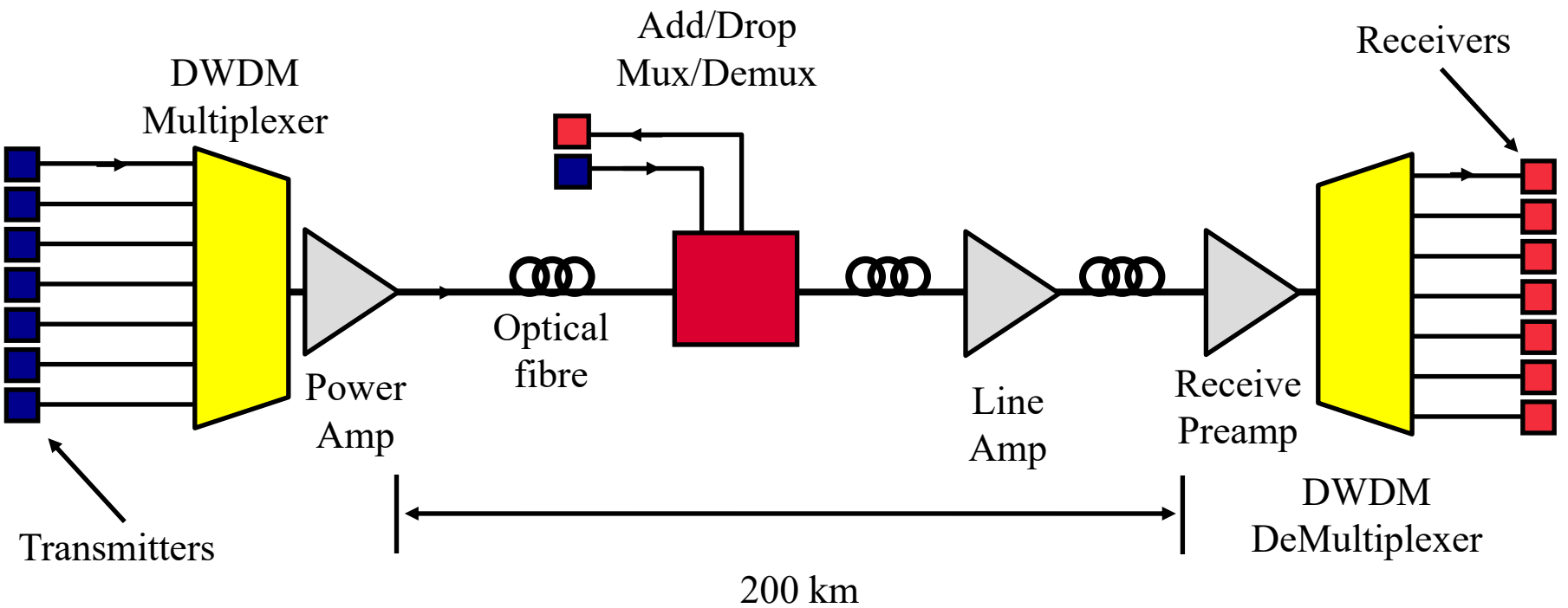
DWDM Systems

DWDM System



- Each wavelength behaves as if it has it own "virtual fibre“.
- Optical amplifiers needed to overcome losses in mux/demux and long fibre spans.

DWDM System with Add-Drop



- Each wavelength still behaves as if it has it own "virtual fibre“.
- Wavelengths can be added and dropped as required at some intermediate location.

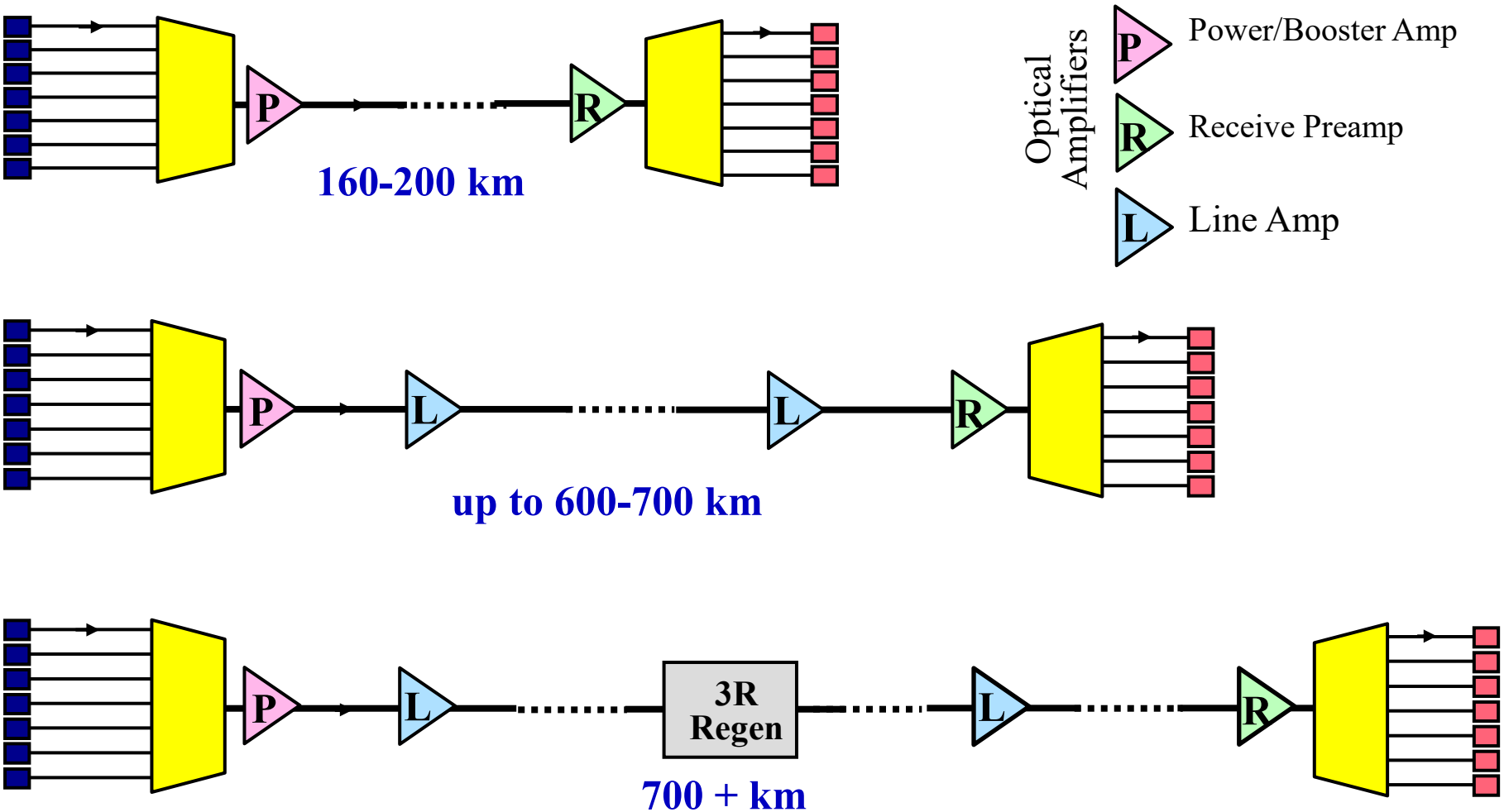
Typical DWDM Systems

Manufacturer & System	Number of Channels	Channel Spacing	Channel Speeds	Maximum Bit Rate Tb/s
Nortel OPtera 1600 OLS	160	0.4 nm	2.5 or 10 Gb/s	1.6 Tbs/s
Lucent	40		2.5	
Alcatel				
Marconi PLT40/80/160	40/80/160	0.4, 0.8 nm	2.5 or 10 Gb/s	1.6 Tb/s

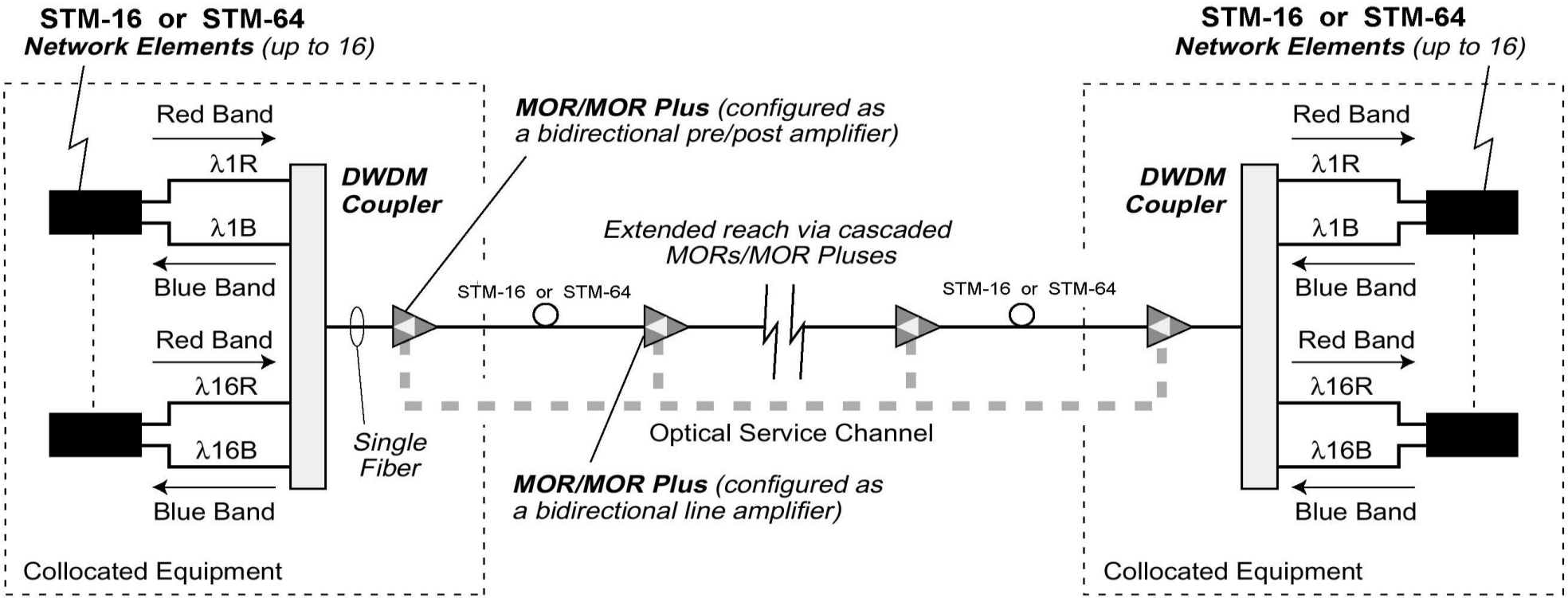
DWDM Performances

- Different systems suit national and metropolitian networks.
- Typical high-end systems currently provide:
  - 40/80/160 channels
  - Bit rates to 10 Gb/s with some 40 Gb/s
  - Interfaces for SDH, PDH, ATM etc.
  - Total capacity to 10 Tb/s +
  - C + L and some S band operation
- Systems available from NEC, Lucent, Marconi, Nortel, Alcatel, Siemens etc.

DWDM System Spans



Nortel DWDM

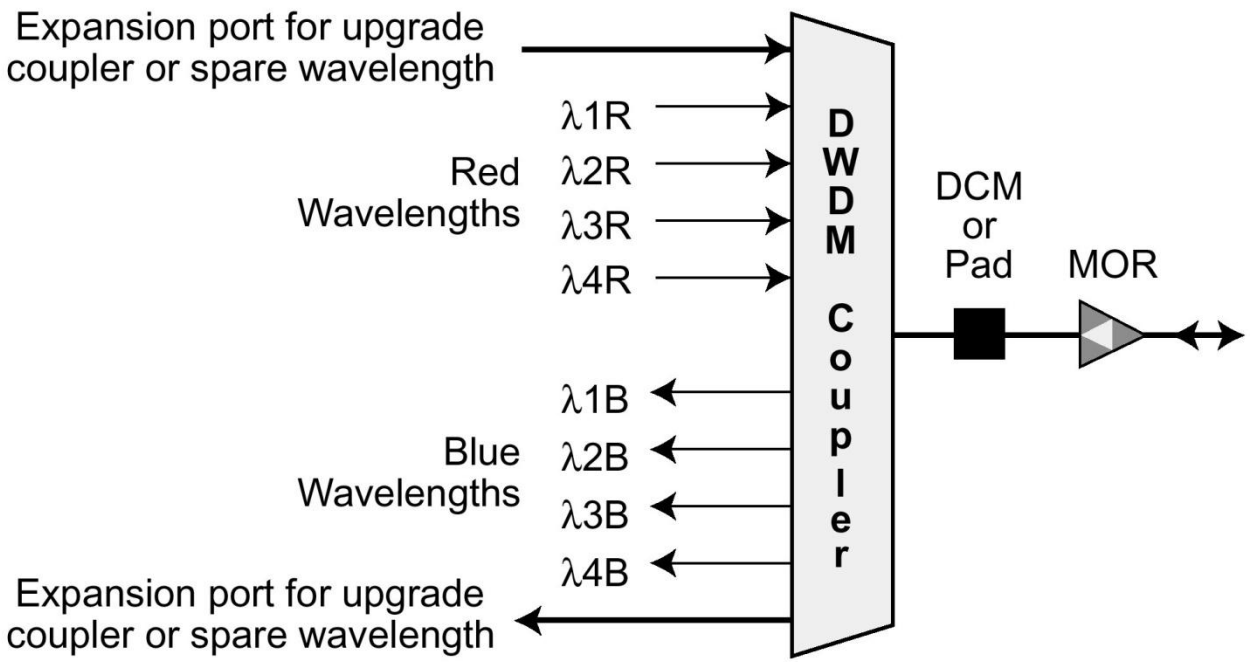


Nortel S/DMS Transport System

- Aggregate span capacities up to 320 Gbits/sec (160 Gbits/sec per direction) possible.
- Red band = 1547.5 to 1561 nm, blue band = 1527.5 to 1542.5 nm.

Nortel DWDM Coupler

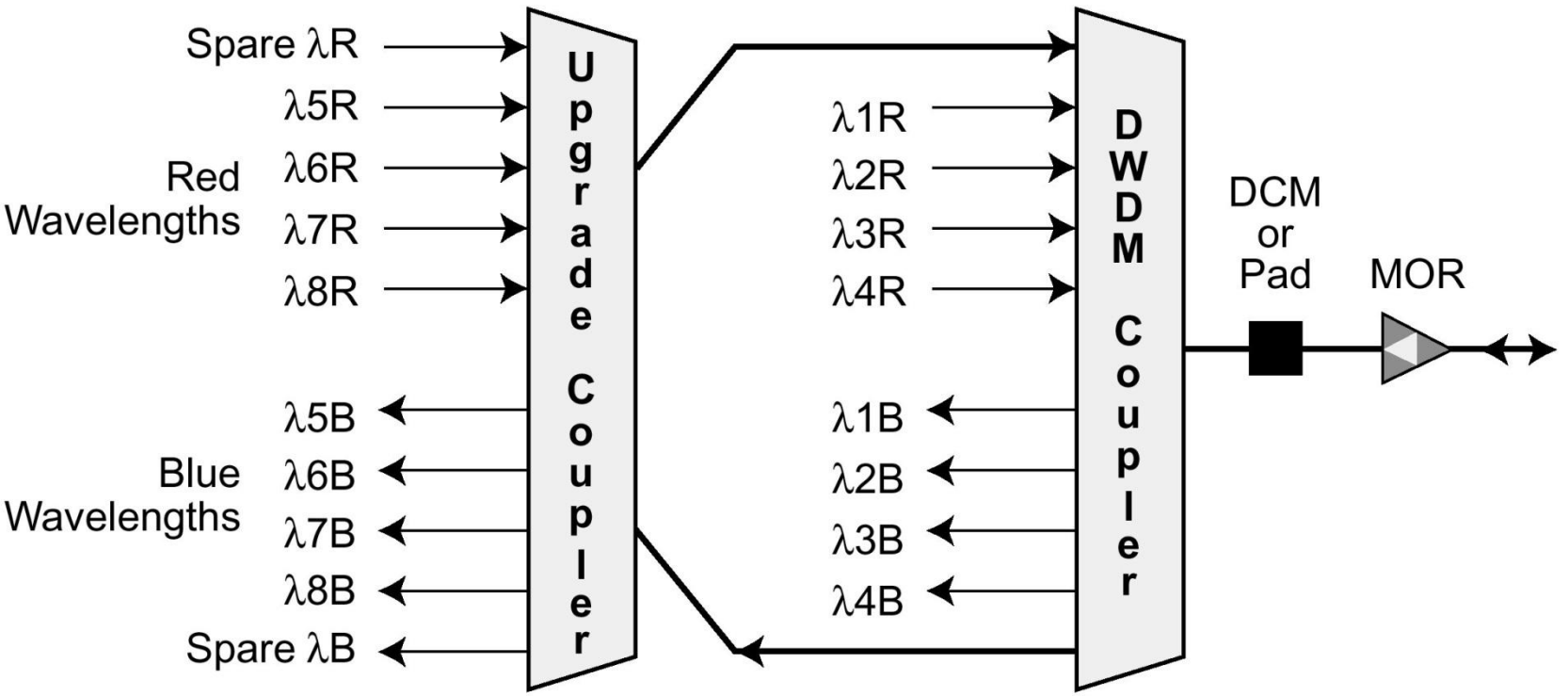
- 8 wavelengths used (4 in each direction) 200 GHz frequency spacing.
- Incorporates a Dispersion Compensation Module (DCM).
- Expansion ports available to allow denser multiplexing.



Red band = 1547.5 to 1561 nm, blue band = 1527.5 to 1542.5 nm

Sixteen Channel Multiplexing

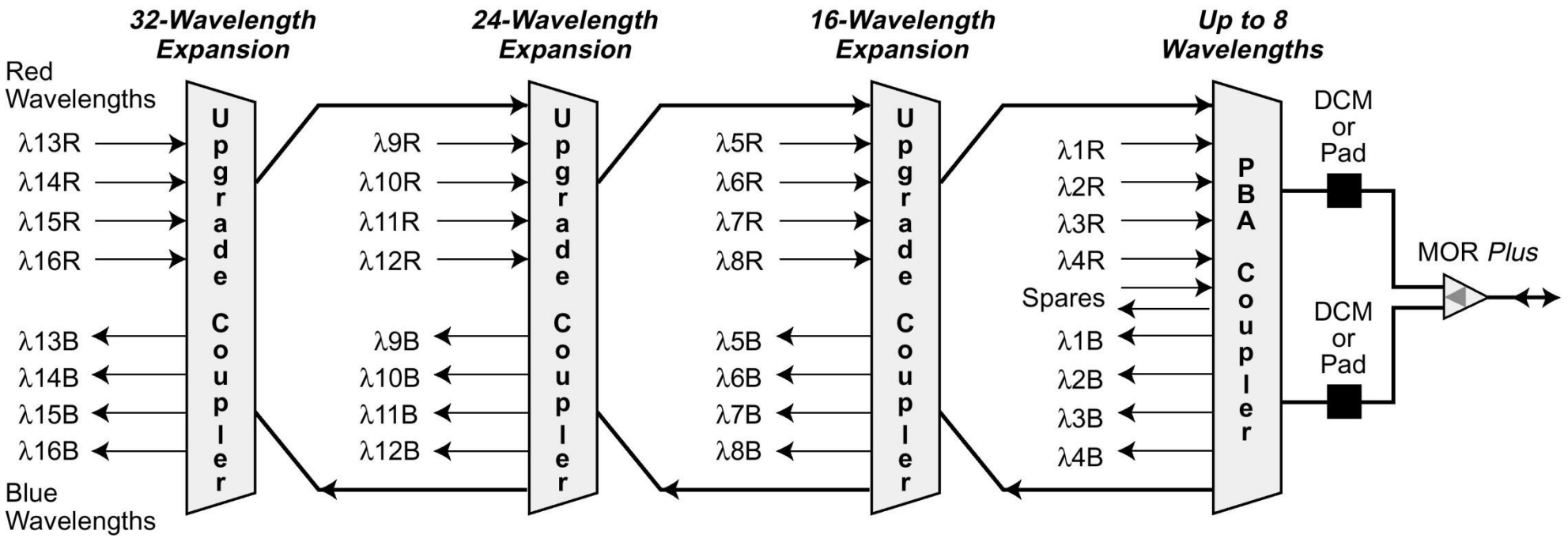
- 16 wavelengths used (8 in each direction), Remains 200 GHz frequency spacing.
- Further expansion ports available to allow even denser multiplexing.



Red band = 1547.5 to 1561 nm, blue band = 1527.5 to 1542.5 nm

32 Channel Multiplexing

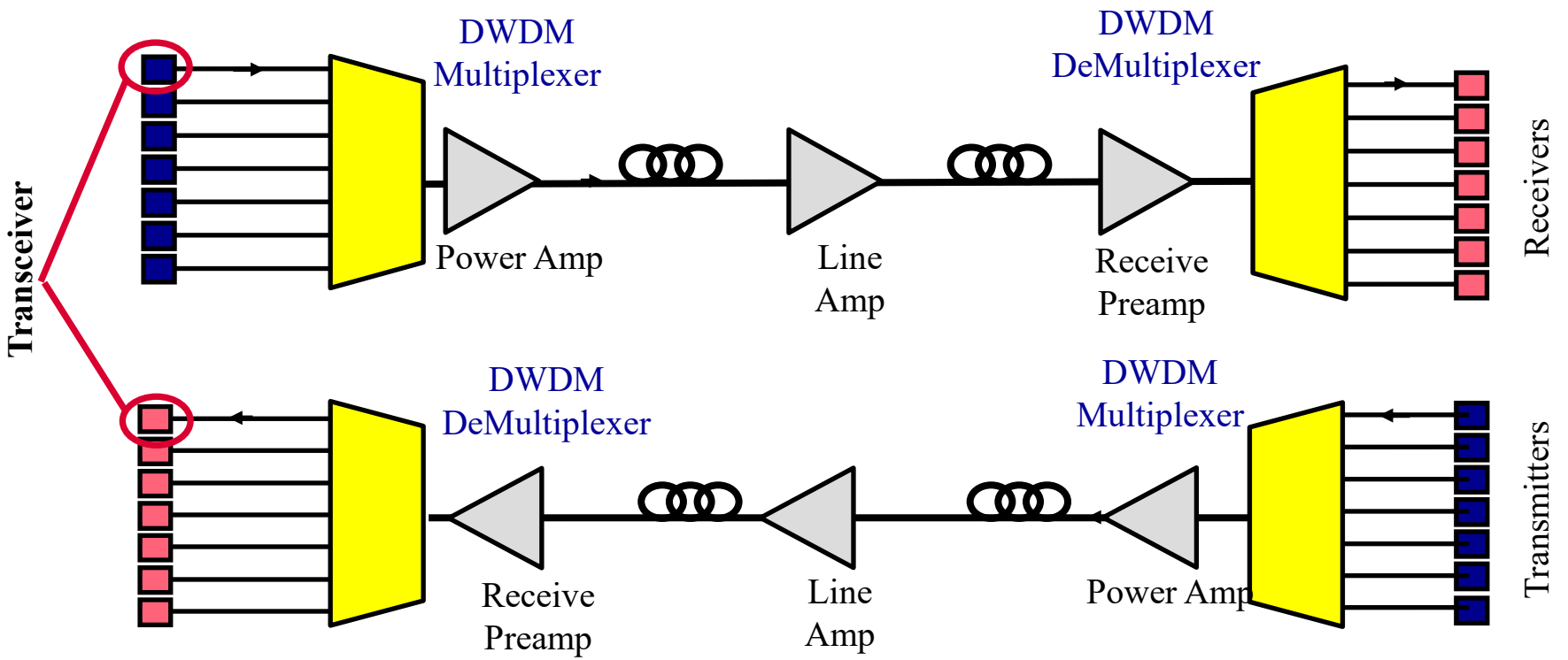
- 32 wavelengths used (16 in each direction) 100 GHz ITU frequency spacing.
- Per band dispersion compensation.



Red band = 1547.5 to 1561 nm, blue band = 1527.5 to 1542.5 nm

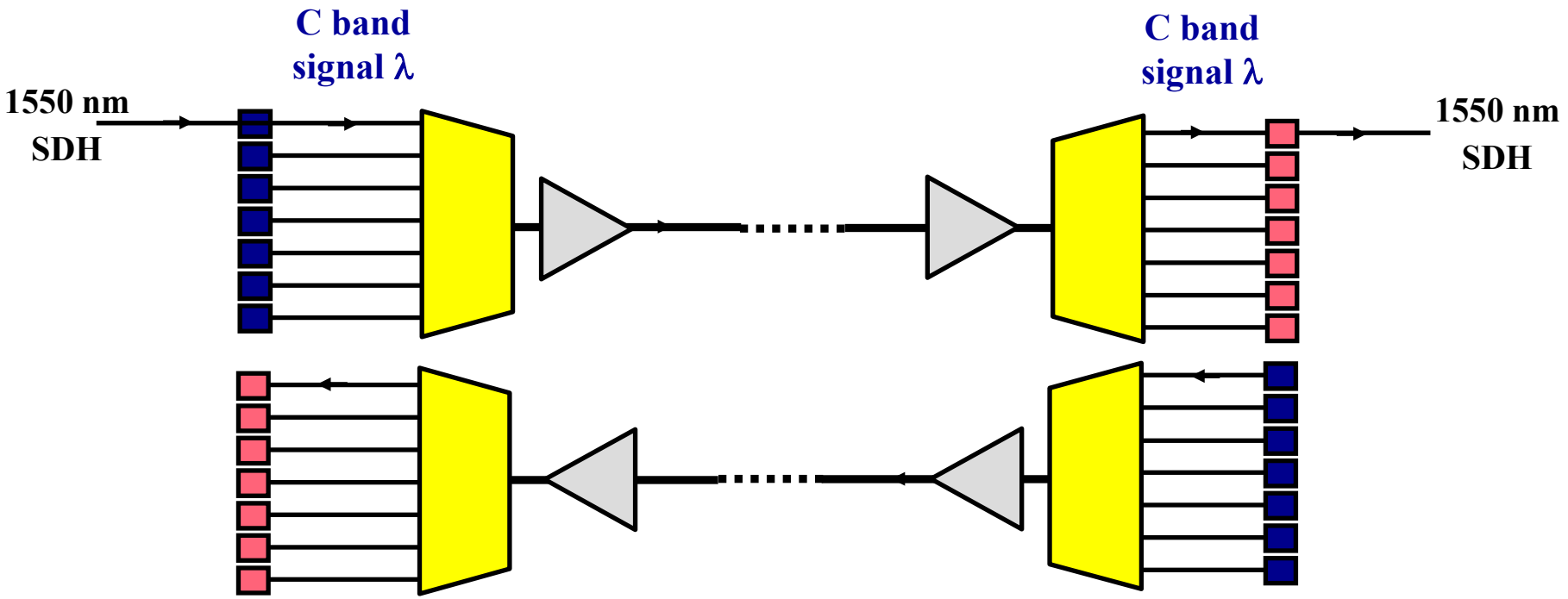
DWDM Transceivers and Transponders

DWDM Transceivers



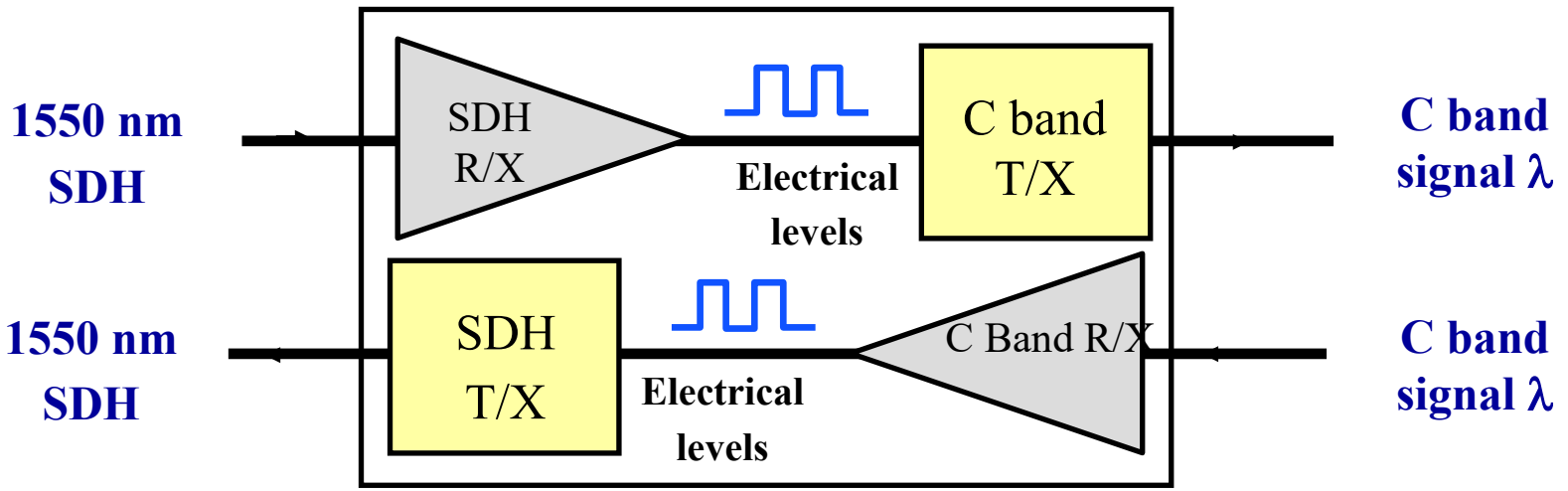
- Transmission in both directions needed.
- In practice each end has transmitters and receivers.
- Combination of transmitter and receiver for a particular wavelength is a "transceiver“.

Transceivers VS Transponders



- In a "classic" system inputs/outputs to/from transceivers are electrical.
- In practice inputs/outputs are SDH, so they are optical, wavelength around 1550 nm.
- In effect we need wavelength convertors not transceivers.
- Such convertors are called transponders.

DWDM Transponders (I)

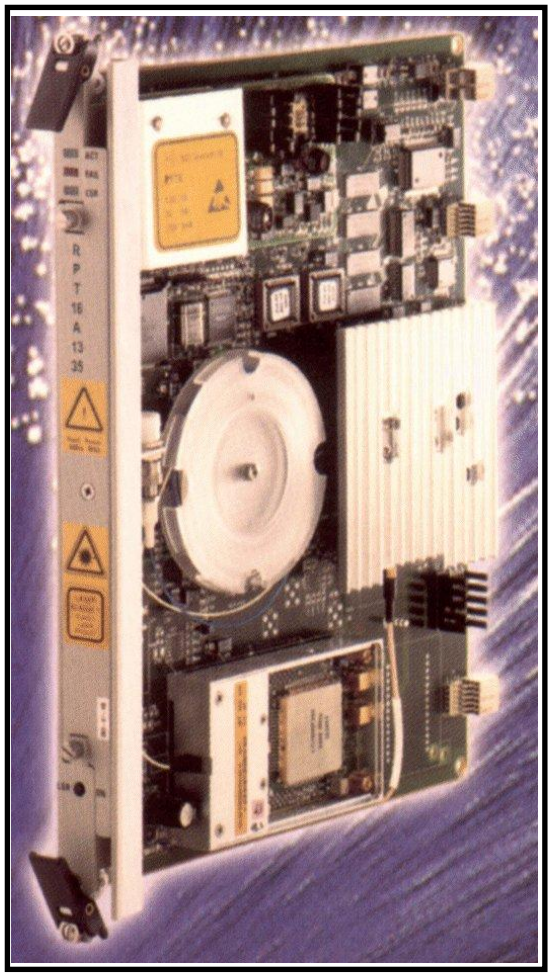


- Transponders are frequently formed by two transceivers back-to-back.
- So called Optical-Electrical-Optical (OEO) transponders.
- Expensive solution at present.
- True all-optical transponders without OEO in development.

DWDM Transponders (II)

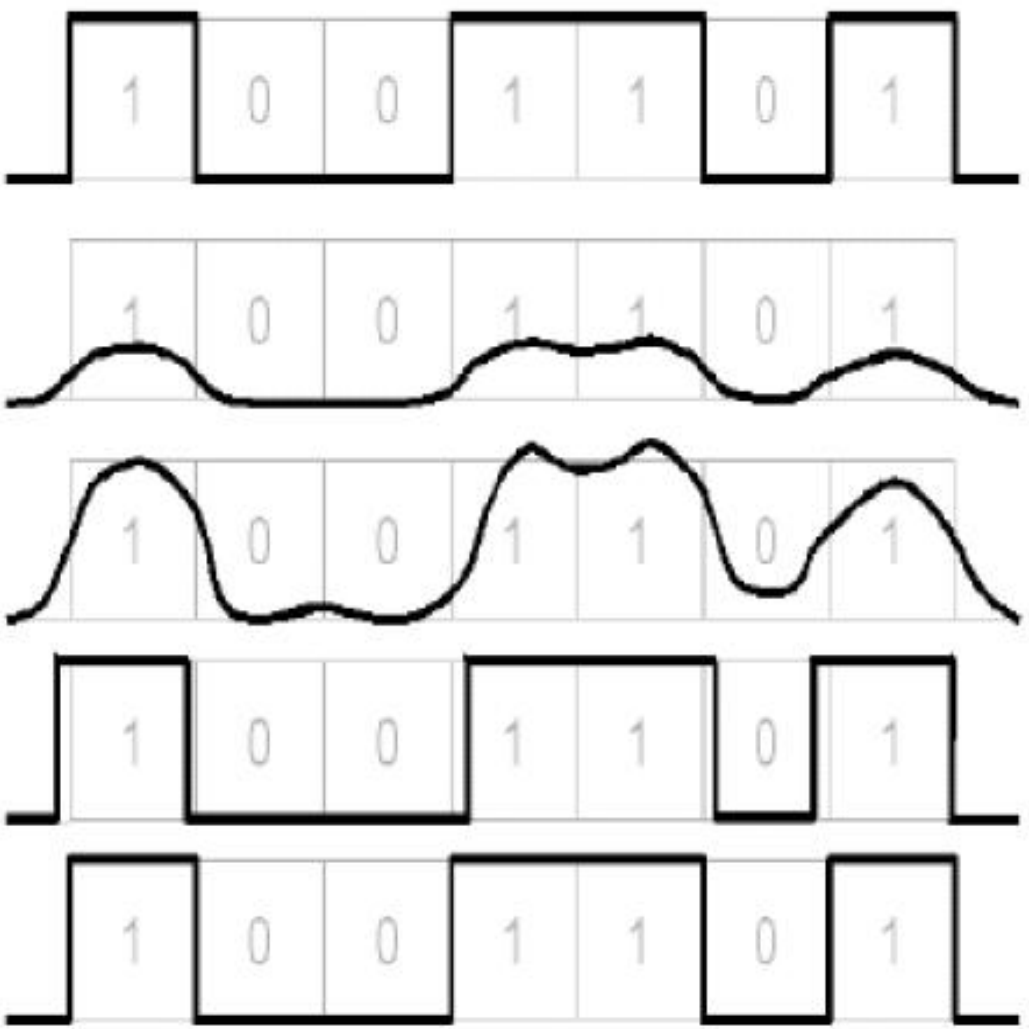
- Full 3R transponders: (power, shape and time)
  - Regenerate data clock
  - Bit rate specific
  - More sensitive - longer range
- 2R transponders also available: (power, shape)
  - Bit rate flexible
  - Less electronics
  - Less sensitive - shorter range

Input Signal	
Wavelength	1260–1360 nm, 1430–1580 nm
3R transponder bit rate input	STM1 / OC3 STM4 / OC12 STM16 / OC48 Gbit Enternet
3R transponder input power	G.957 L-4.1 L-4.2 L-16.1 L-16.2
2R transponder min bit rate input	34 Mbit/s
2R transponder max bit rate input	2.5 Gbit/s
2R transponder input power	G.957 S-1.2 S-4.2 S-16.1 S-16.2



Luminet DWDM Transponder

Amplifiers and Regeneration



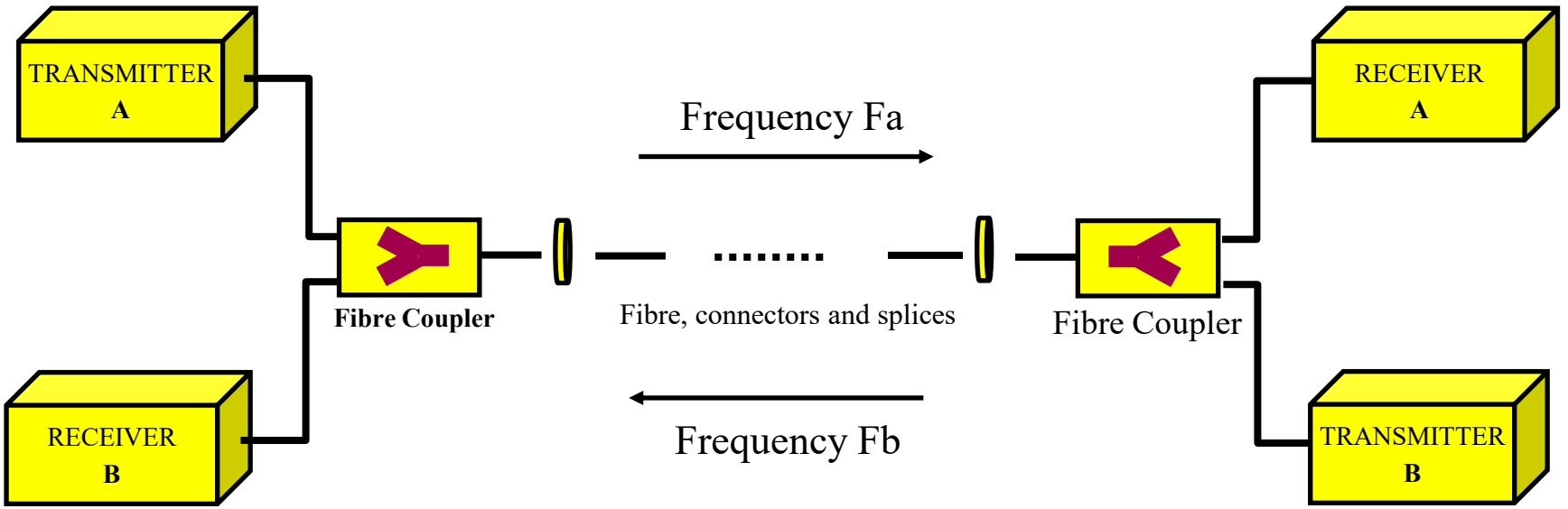
1R—Reamplify

2R—Reamplify and reshape

3R—Reamplify, reshape, and retime

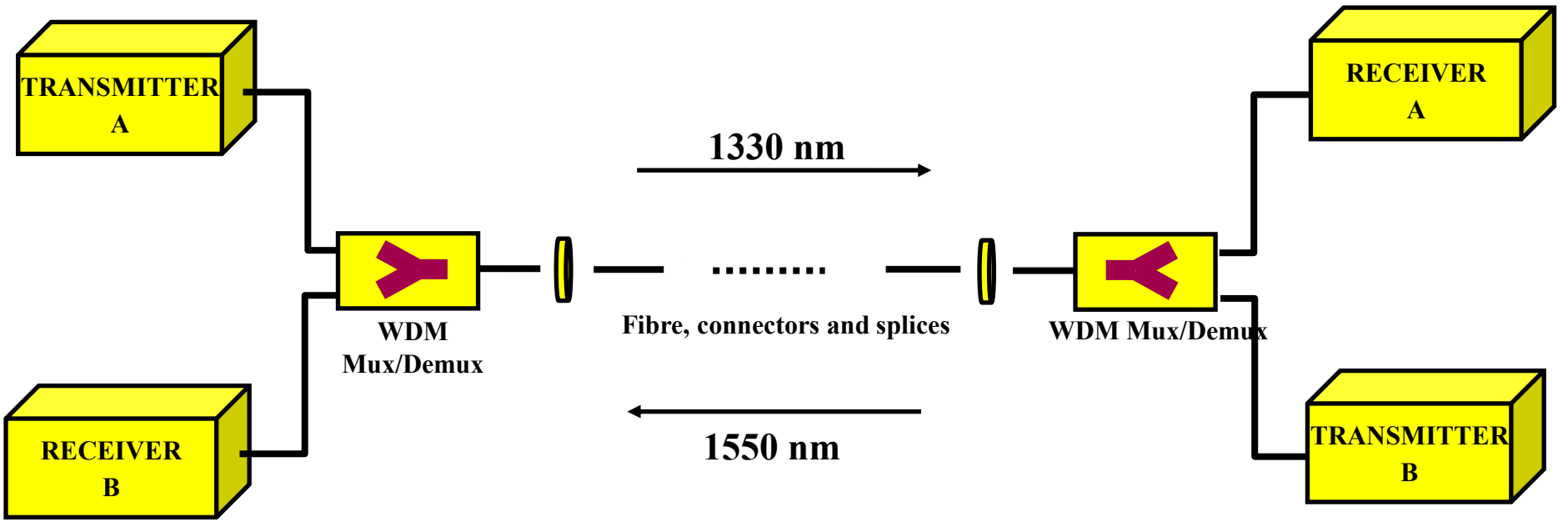
Bidirectional Transmission using WDM

Bi-directional Transmission using Frequency Division Multiplexing



- Transmitter A communicates with Receiver A using a signal on frequency Fa.
- Transmitter B communicates with Receiver B using a signal on frequency Fb.
- Each receiver ignores signals at other frequencies, so for example Receiver A ignores the signal on frequency Fb.

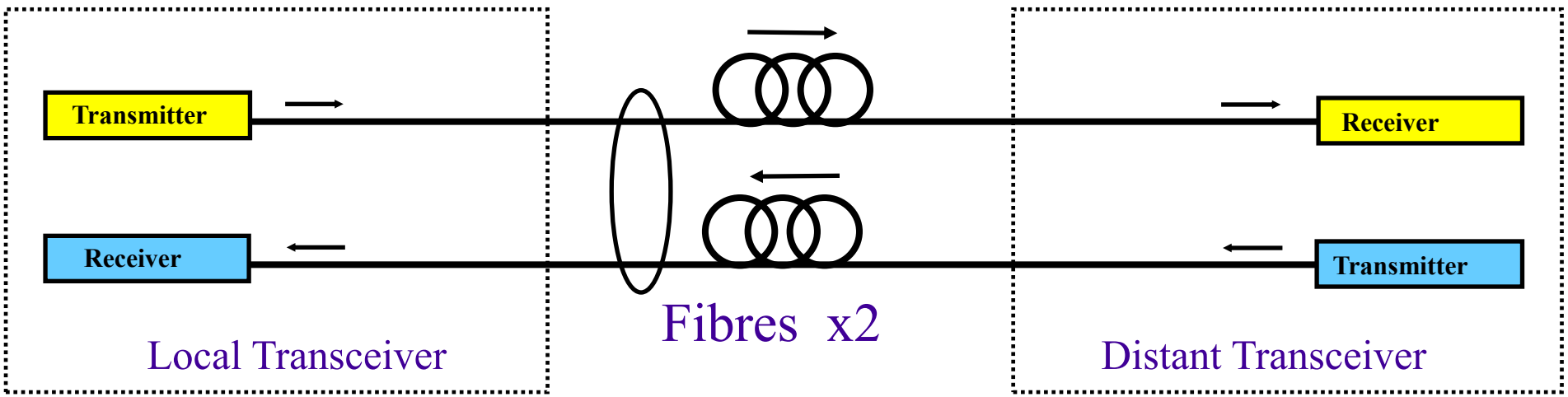
Bi-directional Transmission using WDM



- Transmitter A communicates with Receiver A using a signal on 1330 nm.
- Transmitter B communicates with Receiver B using a signal on 1550 nm.
- WDM Mux/Demux filters out the wanted wavelength so that for example Receiver A only receives a 1330 nm signal.

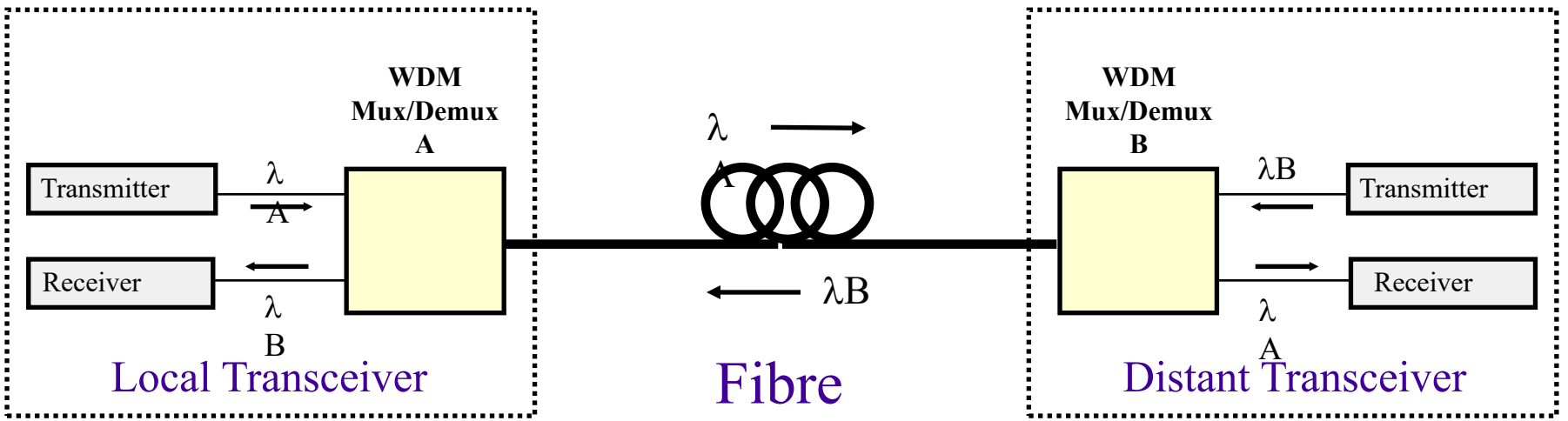
Conventional (Simplex) Transmission

- Most common approach is "one fibre / one direction“.
- This is called "simplex" transmission.
- Linking two locations will involve two fibres and two transceivers.



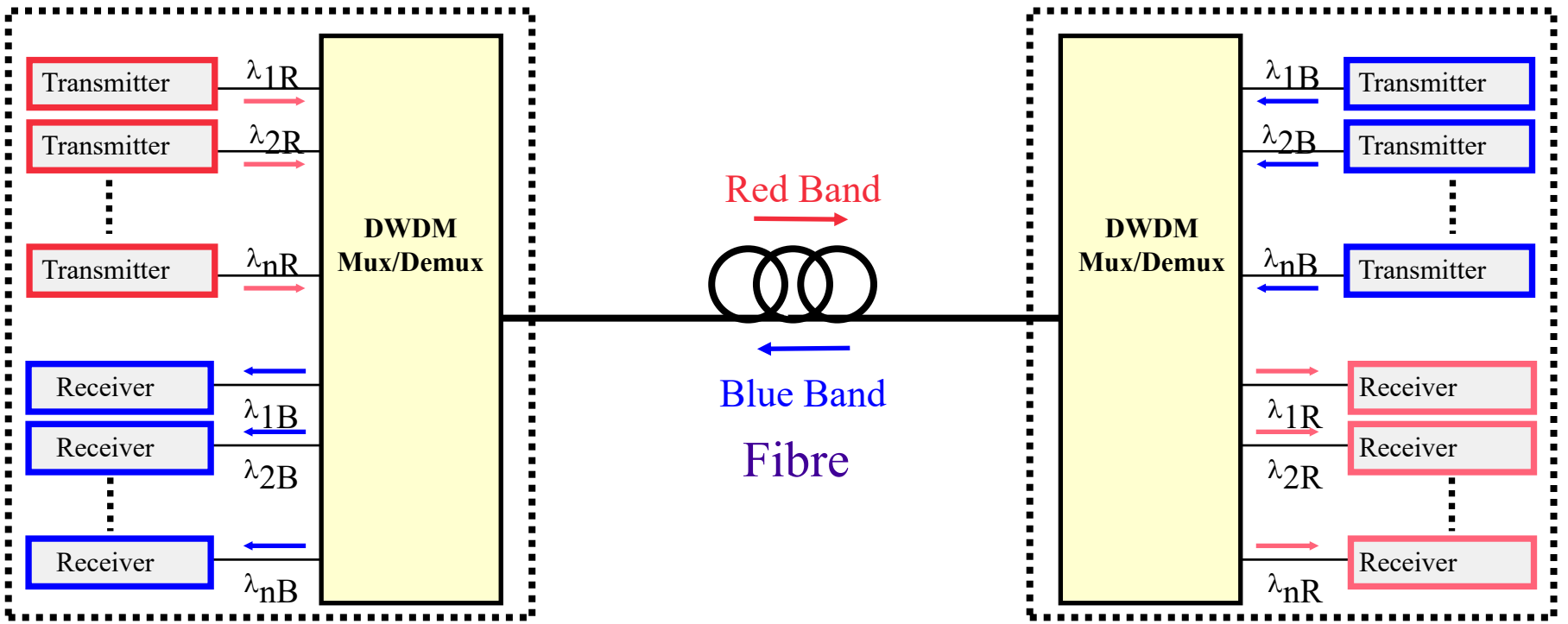
Bi-directional using WDM

- Significant savings possible with so called bi-directional transmission using WDM.
- This is called "full-duplex" transmission.
- Individual wavelengths used for each direction.
- Linking two locations will involve only one fibres, two WDM mux/demuxs and two transceivers.



Bi-directional DWDM

- Different wavelength bands are used for transmission in each direction.
- Typically the bands are called:
  - The "Red Band", upper half of the C-band to 1560 nm
  - The "Blue Band", lower half of the C-band from 1528 nm



The need for a Guard Band

- To avoid interference red and blue bands must be separated.
- This separation is called a "guard band".
- Guard band is typically about 5 nm.
- Guard band wastes spectral space, disadvantage of bi-directional DWDM.

